Model- and Data-Based Approaches to Wind Compensation for Airship Drones

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Abstract

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Acknowledgements

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Nomenclature

Abbreviations

COG	Center of Gravity
COV	Center of Volume
CS	Coordinate System
DFF	Disturbance Feedforward
EKF	Extended Kalman Filter
KF	Kalman Filter
NDI	Nonlinear Dynamic Inversion
NED	North-East-Down convention
UKF	Unscented Kalman Filter

Latin Letters

$\boldsymbol{I}$	Inertia Matrix with respect to the COG

Greek Letters

ω	Angular velocity ($\mathrm{rad\,s^{-1}}$)

Indices

$(\cdot)_B$	Quantity describing the body
$_B(\cdot)$	Quantity in body-fixed coordinate system
$_E(\cdot)$	Quantity in Earth coordinate system

1. Introduction

This chapter serves as an overview of existing methods and ...

1.1. Problem setting

- Why wind is a challenge for airships - Why onboard wind estimation is necessary - Why compensation improves control performance

Atmospheric wind is a major source of disturbance for lighter-than-air vehicles and therefore plays a central role in the development and evaluation of wind estimation and compensation algorithms.

Airships are characterized by their large volume, low flight speed and high aerodynamic drag, making them considerably more susceptible to atmospheric disturbances than conventional fixed-wing aircraft. Even moderate wind speeds may become comparable to the vehicle velocity, causing significant deviations from the desired trajectory and degrading the overall control performance. This sensitivity represents one of the major challenges for autonomous airship operation, particularly during low-speed maneuvers such as hovering, station keeping, inspection tasks and precision landing.

Since the aerodynamic forces and moments acting on an airship depend on the relative velocity between the vehicle and the surrounding air, the ambient wind directly influences the vehicle dynamics. However, wind cannot usually be measured accurately using onboard sensors alone. Installing dedicated wind measurement systems increases the system complexity, weight and cost and is often impractical for small autonomous airships. Consequently, the wind must be estimated from the available onboard measurements together with a mathematical model of the vehicle dynamics.

Reliable wind estimation enables more than merely observing the surrounding atmosphere. By incorporating the estimated wind into the flight controller, the aerodynamic disturbances can be compensated before they lead to significant tracking errors. Such a model-based compensation has the potential to improve trajectory tracking, reduce control effort and increase the robustness of autonomous flight under varying environmental conditions.

Achieving accurate wind estimation and effective disturbance compensation, however, is challenging. The vehicle dynamics are highly nonlinear and are governed by the interaction of gravitational, buoyancy, propulsion and aerodynamic forces, while the aerodynamic effects themselves depend on the unknown wind. Consequently, both the estimator and the controller must account for the nonlinear system dynamics and the coupling between vehicle motion and the surrounding airflow.

1.2. Aim of this work

- Objectives - Scope - Evaluation scenarios

The aim of this work is to develop and implement a model-based algorithm for wind estimation and wind compensation for an unmanned airship. The proposed approach shall reliably estimate the ambient wind using only the available onboard measurements and the nonlinear dynamic model of the vehicle, without requiring dedicated wind sensors.

The wind estimation algorithm is designed to operate under conditions representative of realistic flight scenarios. These include dynamic wind conditions with gusts of up to 30 km/h, as well as different flight maneuvers such as forward and sideward flight, rotation about the vertical axis, and coordinated turning during forward flight. The estimated wind shall remain accurate and robust despite the nonlinear vehicle dynamics and changing operating conditions.

Based on the estimated wind, a model-based compensation strategy is implemented to directly counteract the aerodynamic disturbances acting on the airship. By incorporating the estimated wind into the control system, the influence of wind on the vehicle motion is reduced as directly as possible, thereby improving trajectory tracking performance and the overall robustness of the flight controller.

To evaluate the proposed approach, the complete estimation and compensation framework is implemented in a MATLAB/Simulink simulation environment using a comprehensive nonlinear airship model. The performance of the estimator and the effectiveness of the compensation strategy are assessed under various wind conditions and flight maneuvers representative of realistic operating scenarios.

NDI has been widely applied in flight control, where a single nonlinear controller can provide consistent handling across large flight envelopes without extensive gain-scheduling making it a natural candidate for airship control given the vehicle's wide range of operating speeds, attitudes, and wind conditions.

warum Kalman Filter?

1.3. State of the art / Related work

- Airship dynamics - Wind estimation - Wind compensation / disturbance observers - Positioning of your contribution

1.4. Airship squid

This work is performed on the airship squid, which is further specified in this section. The airship was designed and built by roboloon and consists of a elastic hull, which is usually filled with Helium. For maneuvering the airships has one stern propeller at the back, four fins and four impellers integrated in the fins totalling 9 control surfaces making the airship an overactuated system.

Picture of airship

A full table showing the specifications of the airship can be found in Table A.1

1.4.1. Hardware

The airship is equipped with an IMU, GPS and a 1D pitot tube measuring the airspeed in the body-fixes x axis. This pitot tube yields the restriction of only measuring airspeeds abve 1.5m/s, below this threshold no measurement is available.

1.4.2. Software

The airship is controlled by a PID controller ... The PID is tuned to allow for robust behavior in all flight envelopes.

All algorithms are first implemented in simulation. This simulation poses a model of the airship as derived in Chapter and an implementation of the PID controller. Additionally uncertainties for the forces are considered. After successful validation in the simulation a real world flight test is carried out. ref

Allocator

1.5. Thesis Structure

2. Fundamentals

This chapter introduces the theoretical foundations and key concepts required for the remainder of this thesis. It presents the coordinate systems used throughout this work, the mathematical models of the airship and the wind, the fundamentals of Kalman filtering, and the control concepts underlying the proposed wind compensation approach.

2.1. Coordinate Systems

To describe the motion and dynamics of the airship, as well as the wind velocity acting on the vehicle, two coordinate systems (CS) are considered throughout this work: the body-fixed coordinate system and the earth-fixed coordinate system. Their definitions and the corresponding coordinate transformations are introduced in the following [**Fichter2009Flugmechanik**].

2.1.1. Body-Fixed Coordinate System

The body-fixed coordinate system B is attached to the airship and therefore translates and rotates together with the vehicle. Its origin is located at the Center of Gravity (COG) of the airship, and the coordinate axes are aligned with the body axes according to the North-East-Down (NED) convention. Consequently, the x_B-axis points towards the nose of the airship, the y_B-axis points towards the right side, and the z_B-axis points downwards.

The body-fixed coordinate system is used to describe quantities directly related to the vehicle dynamics, such as translational and rotational velocities, aerodynamic forces, propulsion forces, and moments. Since the equations of motion are formulated in the body-fixed frame, this coordinate system represents the primary reference frame for the dynamic model. A quantity expressed in the body-fixed coordinate system is indicated by a preceding frame identifier, denoted as ${}_B(\cdot)$.

2.1.2. Earth-Fixed Coordinate System

The earth-fixed coordinate system E is introduced as an inertial reference frame for describing the position and orientation of the airship, as well as the wind velocity. For simplicity, a flat-earth assumption is made, which neglects the curvature and rotation of the Earth. This assumption is justified by the relatively short flight distances considered in this work.

The origin of the earth-fixed coordinate system is chosen as an arbitrary point on the Earth's surface. The axes are aligned according to the NED convention, where the north and east axes span the horizontal plane and the down axis points towards the Earth's center. Since the earth-fixed coordinate system is assumed to be inertial, it is considered non-accelerating and non-rotating. A quantity expressed in the earth-fixed coordinate system is indicated by a preceding frame identifier, denoted as ${}_E(\cdot)$.

The transformation from the earth-fixed coordinate system to the body-fixed coordinate system is described by the rotation matrix [1]

$$\boldsymbol{R}_{BE} = \begin{bmatrix} c_\psi c_\theta & s_\psi c_\theta & -s_\theta \\ c_\psi s_\theta s_\phi - s_\psi c_\phi & s_\psi s_\theta s_\phi + c_\psi c_\phi & c_\theta s_\phi \\ c_\psi s_\theta c_\phi + s_\psi s_\phi & s_\psi s_\theta c_\phi - c_\psi s_\phi & c_\theta c_\phi \end{bmatrix}, \tag{2.1}$$

which maps vectors expressed in the earth-fixed coordinate system into the body-fixed coordinate system. $s_{(\cdot)}$ and $c_{(\cdot)}$, denote $\sin(\cdot)$ and $\cos(\cdot)$, respectively.

The inverse transformation is given by the transpose of the rotation matrix

$$\boldsymbol{R}_{EB} = (\boldsymbol{R}_{BE})^T \tag{2.2}$$

and describes the transformation from body-fixed CS to earth-fixed CS.

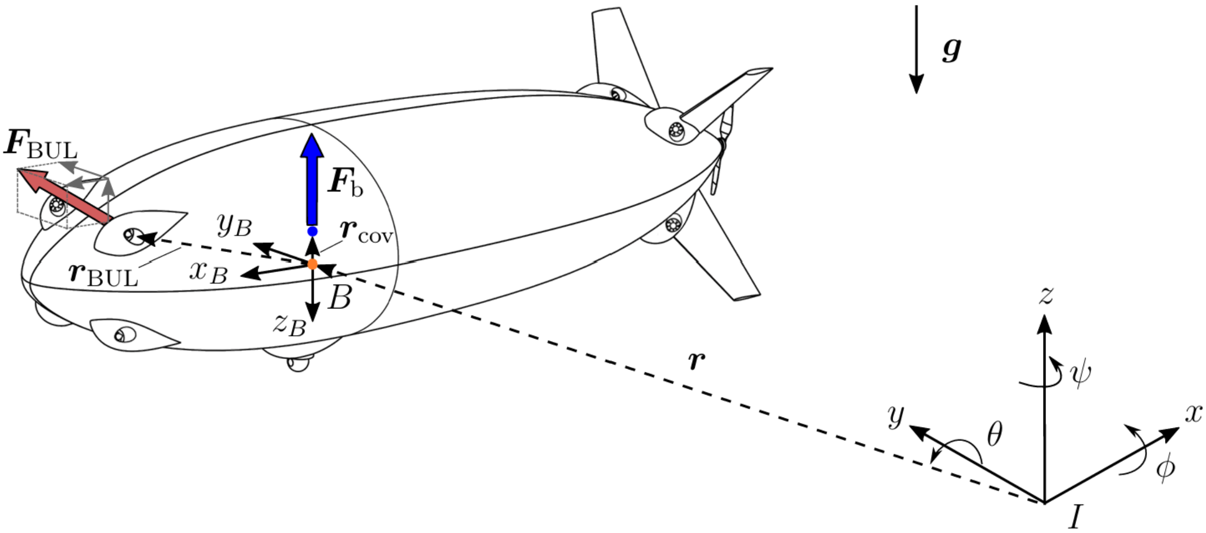

Figure 2.1.: Definition of the earth-fixed and body-fixed coordinate systems. In blue the COV, orange the COG.

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2.2. Airship Model

The airship is modeled using a six degree-of-freedom (6DOF) Newton Euler approach [1]. Therefore we assume the airship to be a rigid body with constant mass and model the translational and rotational dynamics accordingly. We express the equations of motion in body-coordinates, allowing for a constant inertia matrix. Unless stated otherwise, all quantities are expressed in the body-fixed coordinate frame, and all forces and moments are assumed to act at the center of gravity (COG). The corresponding frame identifier $_B(\cdot)$ and point-of-application identifier $(\cdot)^{\mathrm{COG}}$ are omitted throughout the remainder of this section for improved readability.

The three-dimensional **translational dynamics** in body-coordinates are given by the linear momentum theorem as

$$m\boldsymbol{I}_{3\times 3}\,\boldsymbol{a}_B = \boldsymbol{F}_{\text{ext}}, \tag{2.3}$$

where m denotes the mass of the airship, $\boldsymbol{I}_{3\times 3} \in \mathbb{R}^{3\times 3}$ denotes the identity matrix, $\boldsymbol{a}_B \in \mathbb{R}^3$ describes the acceleration of the body (airship) in body-fixed coordinates and $\boldsymbol{F}_{\text{ext}} \in \mathbb{R}^3$ represents the resultant external force acting at the center of gravity (COG), expressed in body-fixed coordinates.

To relate the body acceleration $\boldsymbol{a}_B$ to the body velocity $\boldsymbol{v}_B \in \mathbb{R}^3$, the rotation of the body-fixed coordinate system relative to the earth-fixed coordinate system must be taken into account. Applying Euler's transport theorem yields

$$\boldsymbol{a}_B = \left(\frac{d}{dt}\boldsymbol{v}_B\right) = \dot{\boldsymbol{v}}_B + (\boldsymbol{\omega}_{EB} \times \boldsymbol{v}_B) = \dot{\boldsymbol{v}}_B + \boldsymbol{a}_{\text{cor}}, \tag{2.4}$$

where $\boldsymbol{a}_{\text{cor}} \in \mathbb{R}^3$ denotes the Coriolis acceleration as a result of the rotating body-fixed CS and $\boldsymbol{\omega}_{EB} \in \mathbb{R}^3$ describes the rotational velocity of the body-fixed CS against the Earth CS. As the Earth CS is inertial this reduces to the angular velocity of the airship along its three axes, $\boldsymbol{\omega}_B \in \mathbb{R}^3$. Inserting Eq. (2.4) in Eq. (2.3) and defining the Coriolis force as

$$\boldsymbol{F}_{\text{cor}} = m\boldsymbol{I}_{3\times 3}\,\boldsymbol{a}_{\text{cor}}, \tag{2.5}$$

yields the dynamics for the translational motion of the airship in body-fixed coordinates

$$m\boldsymbol{I}_{3\times 3}\,\dot{\boldsymbol{v}}_B = \boldsymbol{F}_{\text{ext}} - \boldsymbol{F}_{\text{cor}}. \tag{2.6}$$

The three-dimensional **rotational dynamics** in body-fixed coordinates are given by the angular momentum theorem as

$$\left(\frac{d}{dt}\left(\boldsymbol{I}_B\,\boldsymbol{\omega}_B\right)\right) = \boldsymbol{M}_{\text{ext}}, \tag{2.7}$$

where $\boldsymbol{I}_B \in \mathbb{R}^{3\times 3}$ denotes the inertia matrix of the airship expressed in body-fixed coordinates and $\boldsymbol{M}_{\text{ext}} \in \mathbb{R}^3$ represents the resultant external moment acting about the center of gravity (COG), expressed in body-fixed coordinates.

To relate the change of angular momentum to the angular acceleration of the airship, the rotation of the body-fixed coordinate system relative to the earth-fixed coordinate system must be considered. Applying Euler's transport theorem to the angular momentum vector yields

$$\left(\frac{d}{dt}\left(\boldsymbol{I}_B\,\boldsymbol{\omega}_B\right)\right) = \boldsymbol{I}_B\,\dot{\boldsymbol{\omega}}_B + \left(\boldsymbol{\omega}_B \times \boldsymbol{I}_B\,\boldsymbol{\omega}_B\right), \tag{2.8}$$

where the second term represents the gyroscopic acceleration caused by the rotation of the body-fixed coordinate system. Since the inertia matrix is constant when expressed in body-fixed coordinates, the time derivative only applies to the angular velocity.

Defining the gyroscopic moment as

$$\boldsymbol{M}_{\text{gyro}} = \left(\boldsymbol{\omega}_B \times \boldsymbol{I}_B\,\boldsymbol{\omega}_B\right) \tag{2.9}$$

and rearranging Eq. (2.7) yields the rotational dynamics of the airship in body-fixed coordinates

$$\boldsymbol{I}_B\,\dot{\boldsymbol{\omega}}_B = \boldsymbol{M}_{\text{ext}} - \boldsymbol{M}_{\text{gyro}}. \tag{2.10}$$

We can combine the translational dynamics (2.6) and the rotational dynamics (2.10) into one equation obtaining the Newton Euler Equations for the airship.

$$\boldsymbol{M}\dot{\boldsymbol{\nu}} = \boldsymbol{\mathcal{F}}_{\text{ext}} - \boldsymbol{\mathcal{F}}_{\text{cor}} \tag{2.11}$$

with the generalized velocity vector $\boldsymbol{\nu}$ and mass matrix $\boldsymbol{M}$ defined as

$$\boldsymbol{\nu} = \begin{bmatrix} \boldsymbol{v}_B \\ \boldsymbol{\omega}_B \end{bmatrix} \in \mathbb{R}^6, \qquad \boldsymbol{M} = \begin{bmatrix} m\boldsymbol{I}_{3\times3} & \boldsymbol{0}_{3\times3} \\ \boldsymbol{0}_{3\times3} & \boldsymbol{I}_B \end{bmatrix} \in \mathbb{R}^{6\times6}. \tag{2.12}$$

The generalized external and Coriolis force vectors are defined as

$$\boldsymbol{\mathcal{F}}_{\text{ext}} = \begin{bmatrix} \boldsymbol{F}_{\text{ext}} \\ \boldsymbol{M}_{\text{ext}} \end{bmatrix} \in \mathbb{R}^6, \qquad \boldsymbol{\mathcal{F}}_{\text{cor}} = \begin{bmatrix} \boldsymbol{F}_{\text{cor}} \\ \boldsymbol{M}_{\text{cor}} \end{bmatrix} \in \mathbb{R}^6. \tag{2.13}$$

The generalized external forces acting on the airship can be further classified based on the available knowledge of the airship system. Since the airship is modeled as a rigid body, the internal deformations and distributed effects of the structure are neglected, which represents a significant simplification of the real system. Consequently, the accuracy of the airship model depends strongly on the precise representation of the external forces and moments acting on the vehicle. To capture the relevant physical behavior of the airship, these contributions must therefore be modeled as accurately as possible.

However, due to the complexity of the real environment and the limitations of the available model knowledge, not all forces and moments acting on the airship can be fully accounted for. The resulting discrepancies between modeled and actual system behavior are considered model uncertainties. These uncertainties are not explicitly modeled within the presented equations of motion but are addressed in the subsequent estimation and control design.

The external forces and moments acting on the airship are divided into four main categories: gravitational, buoyancy, aerodynamic, and propulsion.

$$\boldsymbol{\mathcal{F}}_{\text{ext}} = \boldsymbol{\mathcal{F}}_{\text{grav}} + \boldsymbol{\mathcal{F}}_{\text{buoy}} + \boldsymbol{\mathcal{F}}_{\text{aero}} + \boldsymbol{\mathcal{F}}_{\text{prop}}, \tag{2.14}$$

resulting in the final EOM

$$\boldsymbol{M}\dot{\boldsymbol{\nu}} = \boldsymbol{\mathcal{F}}_{\text{grav}} + \boldsymbol{\mathcal{F}}_{\text{buoy}} + \boldsymbol{\mathcal{F}}_{\text{aero}} + \boldsymbol{\mathcal{F}}_{\text{prop}} - \boldsymbol{\mathcal{F}}_{\text{cor}}. \tag{2.15}$$

The generalized external forces are derived in the following.

2.2.1. Gravitation and Buoyancy

The gravitational force computes to

$$\boldsymbol{F}_{\text{grav}} = m\boldsymbol{g} = m\boldsymbol{R}_{BE}\,{}_E\boldsymbol{g} \qquad \text{with} \qquad {}_E\boldsymbol{g} = \begin{bmatrix} 0 \\ 0 \\ g \end{bmatrix}, \tag{2.16}$$

where ${}_E\boldsymbol{g}$ represents the constant gravity vector in the inertial earth reference frame following the NED convention and $\boldsymbol{R}_{BE}$ denotes the transformation matrix from earth CS to body CS.

As the gravitational force acts at the center of gravity, no gravitational moment about the COG is induced, thus

$$\boldsymbol{\mathcal{F}}_{\text{grav}} = \begin{bmatrix} \boldsymbol{F}_{\text{grav}} \\ \boldsymbol{0}_{3\times 1} \end{bmatrix}. \tag{2.17}$$

The buoyancy depends on the density of the surrounding air ρ_{air} and the lifting gas contained within the hull ρ_{gas} as well as the volume of the airship V. Assuming a homogeneous distribution, the resulting buoyancy force acts at the center of volume (COV) and is given by

$$\boldsymbol{F}_{\text{buoy}}^{\text{COV}} = -V\boldsymbol{g}(\rho_{\text{air}} - \rho_{\text{gas}}). \tag{2.18}$$

In case of the airship *Squid*, the volume is chosen such that $\boldsymbol{F}_{\text{grav}} + \boldsymbol{F}_{\text{buoy}} > \boldsymbol{0}$, resulting in a downward-directed net force that prevents the airship from ascending without external input. To express the buoyancy contribution with respect to the center of gravity, the moment induced by the offset between the COV and COG is considered:

$$\boldsymbol{M}_{\text{buoy}} = \boldsymbol{r}_{\text{COV}} \times \boldsymbol{F}_{\text{buoy}}^{\text{COV}}, \tag{2.19}$$

where $\boldsymbol{r}_{\text{COV}} \in \mathbb{R}^3$ denotes the vector from the COG to COV. Due to the design of the airship, $\boldsymbol{r}_{\text{COV}}$ only contains a component along the z_B-direction. Consequently, the generalized buoyant force expressed at the COG is given by

$$\boldsymbol{\mathcal{F}}_{\text{buoy}} = \begin{bmatrix} \boldsymbol{F}_{\text{buoy}} \\ \boldsymbol{M}_{\text{buoy}} \end{bmatrix} \qquad \text{with} \qquad \boldsymbol{F}_{\text{buoy}} = \boldsymbol{F}_{\text{buoy}}^{\text{COV}}. \tag{2.20}$$

2.2.2. Aerodynamics

The aerodynamic forces and moments play a central role in the airship dynamics, as they directly depend on the aerodynamic velocity. The aerodynamic velocity is defined as

$$\boldsymbol{v}_A = \boldsymbol{v}_B - \boldsymbol{v}_W \tag{2.21}$$

and denotes the relative velocity between the airship and the surrounding air (often called *airspeed*), where $\boldsymbol{v}_W \in \mathbb{R}^3$ corresponds to the wind velocity expressed in body-fixed coordinates. As a result, the aerodynamical forces and moments contain information about the current wind conditions. An accurate aerodynamic model is therefore essential and forms the basis for the wind estimation approach presented in this work.

As described in [2], the aerodynamic forces and moments can be classified into several contributions, including drag, added mass, viscous effects, and fin forces. Since these contributions account for the majority of the aerodynamic effects, they are considered in the present model.

$$\boldsymbol{\mathcal{F}}_{\text{aero}} = \boldsymbol{\mathcal{F}}_{\text{drag}} + \boldsymbol{\mathcal{F}}_{\text{am}} + \boldsymbol{\mathcal{F}}_{\text{visc}} + \boldsymbol{\mathcal{F}}_{\text{fins}} \tag{2.22}$$

A detailed derivation of the individual contributions is omitted here; instead, the resulting expressions are summarized for completeness. An extensive derivation of the aerodynamic forces and moments can be found in [2]. The aerodynamic forces act at the center of volume. Therefore, the offset between the COV and the COG results in an induced moment, which is considered in the aerodynamic moment contribution.

Aerodynamic Drag

The aerodynamic drag force opposes the relative motion between the airship and the surrounding air and therefore acts in the opposite direction of the aerodynamic velocity. In this work, the drag force is modeled independently along each body-fixed axis using a quadratic drag formulation based on the conventional drag equation [3]. The resulting aerodynamic drag force is given by

$$\boldsymbol{F}_{\text{drag}}^{\text{COV}} = -\frac{1}{2}\rho_{\text{air}}\,\text{sgn}(\boldsymbol{v}_A) \odot \boldsymbol{v}_A^2 \odot \begin{bmatrix} c_{\text{D,ax}} \\ c_{\text{D,lat}} \\ c_{\text{D,lat}} \end{bmatrix}, \tag{2.23}$$

where $\odot$ represents the element-wise (Hadamard) product, and $\text{sgn}(\cdot)$ is applied element-wise. The parameters $c_{\text{D,ax}}$ and $c_{\text{D,lat}}$ are the axial and lateral drag coefficients, respectively. Since the airship is assumed to be rotationally symmetric about its longitudinal axis, the same drag coefficient is used for the lateral and vertical directions.

Since the aerodynamic drag force acts at the COV, it must be transformed to the COG to obtain the corresponding generalized aerodynamic force as

$$\boldsymbol{\mathcal{F}}_{\text{drag}} = \begin{bmatrix} \boldsymbol{F}_{\text{drag}} \\ \boldsymbol{M}_{\text{drag}} \end{bmatrix} \quad \text{with} \quad \boldsymbol{M}_{\text{drag}} = \boldsymbol{r}_{\text{COV}} \times \boldsymbol{F}_{\text{drag}}^{\text{COV}}, \quad \boldsymbol{F}_{\text{drag}} = \boldsymbol{F}_{\text{drag}}^{\text{COV}}. \tag{2.24}$$

Added Mass

For an airship, the added mass can be interpreted as the effective mass of the surrounding air that is accelerated together with the vehicle [4]. The associated added-mass forces and moments arise from the inertia of this accelerated air and contribute to the equations of motion in two ways: through a constant added-mass matrix and through a velocity-dependent coupling term:

$$\boldsymbol{\mathcal{F}}_{\text{am}}^{\text{COV}} = -\begin{bmatrix} \boldsymbol{M}_{11} & \boldsymbol{M}_{12} \\ \boldsymbol{M}_{21} & \boldsymbol{M}_{22} \end{bmatrix} \begin{bmatrix} \dot{\boldsymbol{v}}_B \\ \dot{\boldsymbol{\omega}}_B \end{bmatrix} - \begin{bmatrix} \boldsymbol{\omega}_B \times (\boldsymbol{M}_{11}\boldsymbol{v}_A + \boldsymbol{M}_{12}\boldsymbol{\omega}_B) \\ \boldsymbol{v}_A \times (\boldsymbol{M}_{11}\boldsymbol{v}_A + \boldsymbol{M}_{12}\boldsymbol{\omega}_B) + \boldsymbol{\omega}_B \times (\boldsymbol{M}_{21}\boldsymbol{v}_A + \boldsymbol{M}_{22}\boldsymbol{\omega}_B) \end{bmatrix} \tag{2.25}$$

For steady translation ($\boldsymbol{\omega}_B = \boldsymbol{0}$), the coupling term reduces to the classical *Munk moment*, $-\boldsymbol{v}_A \times (\boldsymbol{M}_{11}\boldsymbol{v}_A)$, which is a destabilizing moment characteristic of elongated bodies first described in [**Munk1924Aerodynamic**].

The entries of the added-mass matrix are computed using the analytical expressions given by [2] and are listed in Appendix .

add ref and write M in appendix

Since the generalized added-mass force acts at the COV, it is transformed to the COG as

$$\boldsymbol{\mathcal{F}}_{\text{am}} = \begin{bmatrix} \boldsymbol{F}_{\text{am}} \\ \boldsymbol{M}_{\text{am}}^{\text{COV}} + \boldsymbol{r}_{\text{COV}} \times \boldsymbol{F}_{\text{am}} \end{bmatrix} \quad \text{with} \quad \boldsymbol{F}_{\text{am}} = \boldsymbol{F}_{\text{am}}^{\text{COV}}. \tag{2.26}$$

Viscous Aerodynamic Effects

In contrast to the added-mass forces, which are associated with the acceleration of the surrounding air, the viscous forces and moments originate from the interaction between the airship surface and the surrounding airflow. These effects arise from viscous shear stresses and pressure differences and are described in detail by [3].

The magnitude of the viscous force and moment acting normal to the hull centerline are given by

$$F_{\mathrm{N,visc}}^{\mathrm{COV}} = q_0 \left(-C_{\mathrm{F,hif}} \sin 2\gamma + C_{\mathrm{F,hvf}} \sin^2 \gamma\right) \tag{2.27}$$

$$M_{\mathrm{N,visc}}^{\mathrm{COV}} = q_0 \left(-C_{\mathrm{M,hif}} \sin 2\gamma + C_{\mathrm{M,hvf}} \sin^2 \gamma\right), \tag{2.28}$$

where q_0 describes the dynamic pressure associated by the local aerodynamic velocity $\boldsymbol{v}_{A,0}$ and γ denotes the angle between the hull centerline and $\boldsymbol{v}_{A,0}$:

$$q_0 = \frac{1}{2}\rho_{\mathrm{air}} \left\|\boldsymbol{v}_{A,0}\right\|_2^2 \qquad \text{and} \qquad \gamma = \mathrm{atan2}\Big(\sqrt{v_{A,0}^2 + w_{A,0}^2}/u_{A,0}\Big) . \tag{2.29}$$

Here $\boldsymbol{v}_{A,0} = [u_{A,0},\ v_{A,0},\ w_{A,0}]^T$ denotes the aerodynamic velocity at ε_0, which marks the point along the hull, measured from the nose, beyond which correction for real flow behavior is necessary. The coefficient $C_{\cdot,\mathrm{hif}}$ accounts for the additional drag caused by the pressure difference between the front and rear of the hull after flow separation, whereas $C_{\cdot,\mathrm{hvf}}$ represents the crossflow drag caused by the separated flow, modeled using a cylinder-drag analogy.

The viscous force and moment are then decomposed into their body-axis components. Since both act normal to the hull centerline $F_{\mathrm{visc},x}^{\mathrm{COV}} = M_{\mathrm{visc},x}^{\mathrm{COV}} = 0$ and

$$F_{\mathrm{visc},y}^{\mathrm{COV}} = -F_{\mathrm{N,visc}}^{\mathrm{COV}} \frac{v_{A,0}}{\sqrt{v_{A,0}^2 + w_{A,0}^2}}, \qquad F_{\mathrm{visc},z}^{\mathrm{COV}} = -F_{\mathrm{N,visc}}^{\mathrm{COV}} \frac{w_{A,0}}{\sqrt{v_{A,0}^2 + w_{A,0}^2}} \tag{2.30}$$

and the corresponding body-axis moment components are

$$M_{\mathrm{visc},y}^{\mathrm{COV}} = M_{\mathrm{N,visc}}^{\mathrm{COV}} \frac{v_{A,0}}{\sqrt{v_{A,0}^2 + w_{A,0}^2}}, \qquad M_{\mathrm{visc},z}^{\mathrm{COV}} = -M_{\mathrm{N,visc}}^{\mathrm{COV}} \frac{w_{A,0}}{\sqrt{v_{A,0}^2 + w_{A,0}^2}}. \tag{2.31}$$

Again, we have to transform the force and moment to the COG.

$$\boldsymbol{\mathcal{F}}_{\mathrm{visc}} = \begin{bmatrix} \boldsymbol{F}_{\mathrm{visc}} \\ \boldsymbol{M}_{\mathrm{visc}}^{\mathrm{COV}} + \boldsymbol{r}_{\mathrm{COV}} \times \boldsymbol{F}_{\mathrm{visc}} \end{bmatrix} \qquad \text{with} \qquad \boldsymbol{F}_{\mathrm{visc}} = \boldsymbol{F}_{\mathrm{visc}}^{\mathrm{COV}}. \tag{2.32}$$

Fin Forces

The fin forces are computed using a lumped approach, in which each of the four fins is treated as a single lifting surface. The contribution of each fin depends on the local aerodynamic velocity and the corresponding angle of attack. The total fin force and moment are then obtained by summing the individual fin contributions.

For each fin $i = 1, \dots, 4$, located at position $\boldsymbol{r}_{\text{fin},i}$ relative to the COV, the local aerodynamic velocity follows from the rigid-body velocity field

$$\boldsymbol{v}_{A,\text{fin},i} = \boldsymbol{v}_A + \boldsymbol{\omega}_B \times \boldsymbol{r}_{\text{fin},i}, \tag{2.33}$$

from which the local dynamic pressure is obtained as

$$q_{0,\text{fin},i} = \frac{1}{2}\rho_{\text{air}} \left\| \boldsymbol{v}_{A,\text{fin},i} \right\|_2^2 . \tag{2.34}$$

The geometric angle of attack of each fin is computed from the local velocity component perpendicular to the fin surface $v_{\perp,\text{fin},i}$ and the local velocity component along the hull centerline $u_{\text{fin},i}$

$$\alpha_{\text{fin},i} = \operatorname{atan2}\left(\frac{v_{\perp,\text{fin},i}}{u_{\text{fin},i}}\right), \tag{2.35}$$

and is saturated at the stall angle α_{stall}.

The resulting normal force on each fin computes to

$$F_{N,\text{fin},i} = -q_{0,\text{fin},i}\, C_{\text{F,fnf}}\, \alpha_{\text{fin},i}, \tag{2.36}$$

where $C_{\text{F,fnf}}$ is the empirical fin normal-force coefficient that is assumed to capture all aerodynamic effects of the fins, including geometric and interference effects, in a single fitted parameter.

The total fin force and moment about the COV are obtained by summing the individual fin contributions,

$$\boldsymbol{F}_{\text{fins}}^{\text{COV}} = \sum_{i=1}^{4} \boldsymbol{F}_{\text{fin},i}, \qquad \boldsymbol{M}_{\text{fins}}^{\text{COV}} = \sum_{i=1}^{4} \boldsymbol{r}_{\text{fin},i} \times \boldsymbol{F}_{\text{fin},i}, \tag{2.37}$$

where $\boldsymbol{F}_{\text{fin},i}$ denotes the fin force vector obtained by resolving the fin normal force $F_{N,Fi}$ into its body-fixed components according to the orientation of the respective fin.

As with the previous contributions, the resulting force and moment are transformed to the COG to obtain the generalized fin forces

$$\boldsymbol{\mathcal{F}}_{\text{fins}} = \begin{bmatrix} \boldsymbol{F}_{\text{fins}}^{\text{COV}} \\ \boldsymbol{M}_{\text{fins}}^{\text{COV}} + \boldsymbol{r}_{\text{COV}} \times \boldsymbol{F}_{\text{fins}}^{\text{COV}} \end{bmatrix} \qquad \text{with} \qquad \boldsymbol{F}_{\text{fins}} = \boldsymbol{F}_{\text{fins}}^{\text{COV}}. \tag{2.38}$$

2.2.3. Propulsion

The propulsion forces and moments resulting from the command input $\boldsymbol{u} \in \mathbb{R}^9$ are computed in two steps. First, the thrust generated by each individual motor $F_{\text{motor},i}$ is determined from the corresponding motor command u_i using a nonlinear motor model. Due to the different characteristics of the impellers and the stern propeller, separate parameter sets are used for the two actuator types. The resulting motor thrust is therefore given by

$$F_{\text{motor},i} = \begin{cases} \alpha_{\text{imp}} \left(k_{\text{imp}}\, u_i^2 + (1 - k_{\text{imp}}) u_i\right), & i = 1, \dots, 8 \\ \alpha_{\text{stern}} \operatorname{sgn}(u_i) \left(k_{\text{stern}}\, u_i^2 + (1 - k_{\text{stern}}) u_i\right), & i = 9 \end{cases} \tag{2.39}$$

The individual motor thrusts are then combined into the resulting generalized force vector by considering the orientation and position of each motor. This transformation is described by the motor effectiveness matrix $\boldsymbol{B}_{\text{eff}}^{\text{COG}} \in \mathbb{R}^{6\times 9}$, which maps the individual motor thrusts to the generalized force vector acting at the COG. The matrix incorporates the orientation of each motor as well as the moment contributions resulting from their respective positions relative to the COG.

$$\boldsymbol{\mathcal{F}}_{\text{prop}} = \boldsymbol{B}_{\text{eff}}^{\text{COG}} \boldsymbol{F}_{\text{motor}}, \tag{2.40}$$

with

$$\boldsymbol{B}_{\text{eff}}^{\text{COG}} = \begin{bmatrix} \boldsymbol{d}_1 & \cdots & \boldsymbol{d}_9 \\ \boldsymbol{r}_1 \times \boldsymbol{d}_1 & \cdots & \boldsymbol{r}_9 \times \boldsymbol{d}_9 \end{bmatrix}, \tag{2.41}$$

where $\boldsymbol{r}_i \in \mathbb{R}^3$ denotes the position vector of the i-th motor relative to the COG, and $\boldsymbol{d}_i \in \mathbb{R}^3$ denotes the corresponding unit thrust direction vector.

2.3. Wind Model

To ensure that the simulation results are representative of real flight conditions, the wind model needs to accurately capture the dominant characteristics of atmospheric wind while remaining computationally efficient. In particular, a state-space realization is required so that the wind model can be incorporated into the airship dynamics, providing the necessary states for wind estimation within the Kalman filter framework. Since atmospheric wind generally consists of a slowly varying mean component combined with rapidly changing turbulence, the wind in this work is modeled as the sum of a constant mean wind and a stochastic turbulence component [**Etkin1995Dynamics**].

$$\boldsymbol{v}_W = \boldsymbol{v}_{W,\text{const}} + \boldsymbol{v}_{W,\text{turb}}. \tag{2.42}$$

Stochastic turbulence models are widely used in aerospace applications because they reproduce the statistical properties of atmospheric turbulence without requiring an explicit description of the underlying turbulent wind field. A simple stochastic model is the random walk, in which white noise is integrated to generate a randomly varying signal. Among the most commonly used models are the Dryden and the von Kármán turbulence models. Although the von Kármán model generally provides a more accurate representation of the atmospheric turbulence spectrum, it describes turbulence using non-rational functions and therefore cannot be represented exactly by a finite-dimensional state-space model. In contrast, the Dryden turbulence model approximates atmospheric turbulence by passing white noise through linear shaping filters, resulting in rational transfer functions that are realizable in state-space form [**Hoblit1988Gust**, **MILHDBK1797**].

Due to its ease of implementation while still describing atmospheric turbulence sufficiently well, the Dryden turbulence model is adopted throughout this work. Figure 2.2 compares wind turbulences generated by a random walk model, the Dryden turbulence model and the von Kármán turbulence model. While the random walk exhibits an unrealistic long-term drift, the Dryden model produces bounded, correlated wind disturbances similar to the von Kármán model.

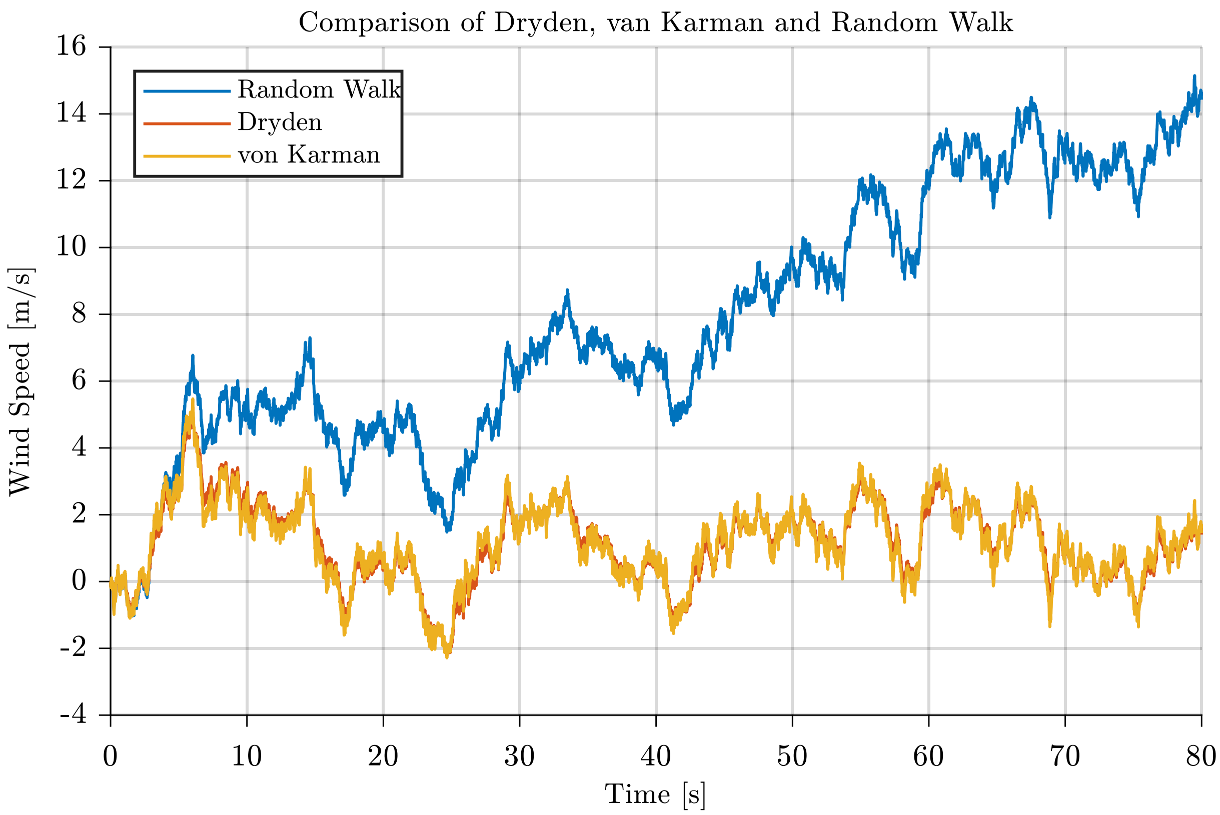


Figure 2.2.: Comparision of wind turbulences created by the random walk, von Kármán and Dryden model for identical white noise input.

2.3.1. Dryden Turbulence Model

The Dryden turbulence model, named after the American aerodynamicist Hugh L. Dryden, is a stochastic model for continuous atmospheric turbulence. The model is based on statistical descriptions of turbulence and was first formulated in terms of power spectral densities by Liepmann in 1952 [**Liepmann1952Dryden**].

Rather than generating turbulent wind directly in the time domain, the Dryden model characterizes atmospheric turbulence by its power spectral density and synthesizes the corresponding wind components by filtering white noise. As illustrated in Figure 2.3, three independent white Gaussian noise processes are passed through shaping filters for the longitudinal, lateral, and vertical directions, resulting in the stochastic turbulence velocity vector $\boldsymbol{v}_{W,\mathrm{turb}} = [u_{W,\mathrm{turb}},\, v_{W,\mathrm{turb}},\, w_{W,\mathrm{turb}}]^T$. A white Gaussian noise process describes a sequence of independent Gaussian random variables indexed in time. Each sample is normally distributed with zero mean and constant variance, while the values at different time instants are uncorrelated.

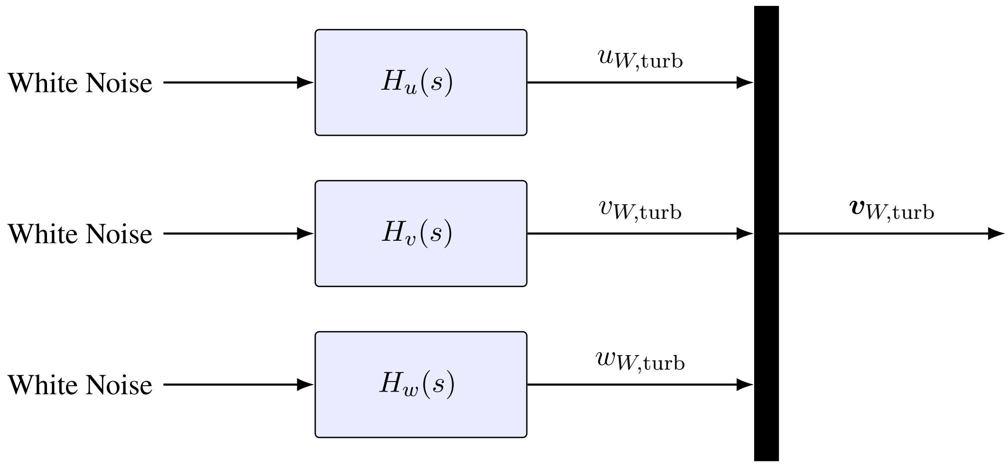


Figure 2.3.: Structure of the Dryden Turbulence model.

For this work, the low-altitude variant of the Dryden model (valid for altitudes below $300\,\mathrm{m}$, as specified in MIL-F-8785C / MIL-HDBK-1797 [**MILF8785C**, **MILHDBK1797**]) is used, consistent with the near-ground operational envelope of the airship. In this formulation, the Dryden shaping filters generate the turbulence components in a coordinate system aligned with the mean wind. The longitudinal component $u_{W,\mathrm{turb}}$ is defined along the horizontal mean wind direction, the lateral component $v_{W,\mathrm{turb}}$ lies perpendicular to it in the horizontal plane, and the vertical component $w_{W,\mathrm{turb}}$ is aligned with the z-axis of the Earth-fixed coordinate system. Consequently, the filter outputs are not expressed in the vehicle's body-fixed coordinate system. The symbols u, v, w are nevertheless maintained because they represent the standard notation for the Dryden turbulence components.

Since no onboard measurement of the true mean wind direction is available and only the total wind, rather than its mean and turbulent components, is estimated, the orientation of the mean-wind coordinate system is unknown. Therefore, for simplicity, the mean-wind coordinate system is assumed to coincide with the Earth-fixed coordinate system, with the mean wind aligned with the positive Earth x-axis. Under this assumption, the turbulence vector $\boldsymbol{v}_{W,\mathrm{turb}}$ is given directly in the Earth-fixed frame.

The corresponding shaping filters, which reproduce the characteristic Dryden turbulence spectra, are defined by the following transfer functions [**MILHDBK1797**]:

$$H_u(s) = \sigma_u \sqrt{\frac{2L_u}{\pi V}} \frac{1}{1 + \frac{L_u}{V} s} \tag{2.43}$$

$$H_v(s) = \sigma_v \sqrt{\frac{L_v}{\pi V}} \frac{1 + \frac{\sqrt{3} L_v}{V} s}{\left(1 + \frac{L_v}{V} s\right)^2} \tag{2.44}$$

$$H_w(s) = \sigma_w \sqrt{\frac{L_w}{\pi V}} \frac{1 + \frac{\sqrt{3} L_w}{V} s}{\left(1 + \frac{L_w}{V} s\right)^2}, \tag{2.45}$$

where $V = \|\boldsymbol{v}_B\|_2$ denotes the absolute velocity of the airship and σ, L are the Dryden turbulence parameters. The longitudinal turbulence component is represented by a first-order transfer function, whereas the lateral and vertical components require second-order transfer functions due to their different spectral characteristics. A detailed derivation of the Dryden transfer functions can be found in [**MILHDBK1797**, **Yeager1998Turbulence**]. It is worth noting that for $V = 0$, such as during hovering, the generated Dryden turbulence computes to zero. The corresponding filter parameters, including the turbulence intensities σ and length scales L, are discussed in the following.

Dryden Model Parameters

The Dryden shaping filters are parameterized by the turbulence intensities σ_u, σ_v, σ_w and the turbulence scale lengths L_u, L_v, and L_w. These parameters determine the strength and spatial correlation of the generated turbulence and are specified in [**MILF8785C**]. Although the original specification defines the parameters using imperial units, the equivalent expressions in SI units are employed throughout this work.

For flight altitudes below $300\,\mathrm{m}$, the turbulence scale lengths are defined as

$$L_w = h, \tag{2.46}$$

$$L_u = L_v = \frac{h}{(0.177 + 0.0027\,h)^{1.2}}, \tag{2.47}$$

where h denotes the altitude above ground in meters. The corresponding turbulence intensities are computed as

$$\sigma_w = 0.1\,W_{20}, \tag{2.48}$$

$$\sigma_u = \sigma_v = \frac{\sigma_w}{(0.177 + 0.0027\,h)^{0.4}}, \tag{2.49}$$

where W_{20} denotes the wind speed at an altitude of $6\,\mathrm{m}$ ($20\,\mathrm{ft}$). Consequently, the characteristics of the generated turbulence are determined by only two parameters: the flight altitude h and the reference wind speed W_{20}.

Since the turbulence intensities σ scale linearly with W_{20}, this parameter directly determines the magnitude of the generated turbulence. Therefore, an increase of W_{20} results in a proportional increase in the generated turbulence intensity. Conversely, for $W_{20} = 0$, the turbulence intensities become zero, and the Dryden model produces no turbulence contribution.

2.4. Kalman Filter

[**Maybeck1979KF**] [**SÃďrkkÃď2013KF**]

The Kalman filter (KF), introduced by Kalman in 1960 [**Kalman1960KF**], is one of the most widely used algorithms for state estimation in dynamic systems. Its objective is to estimate the state of a system from noisy or incomplete sensor measurements by recursively combining an uncertain mathematical model of the system with incoming measurements. At each time step, the filter first predicts the current state based on the system dynamics and then refines this prediction using the latest measurement. The prediction and measurement are weighted according to their respective uncertainties, allowing the filter to balance the information provided by the system model and the sensor measurements.

Under the assumptions of linear system dynamics the Kalman filter yields the optimal state estimate in the minimum mean-square error (MMSE) sense [**Simon2006KF**]. More specifically, at each time step k, it computes the state estimate that minimizes the expected squared estimation error conditioned on all available measurements,

$$\hat{\boldsymbol{x}}_k = \arg\min_{\tilde{\boldsymbol{x}}} \mathbb{E}\left[\|\boldsymbol{x}_k - \tilde{\boldsymbol{x}}\|_2^2 \,\middle|\, \boldsymbol{y}_{1:k} \right], \tag{2.50}$$

where $\boldsymbol{y}_{1:k} = \{\boldsymbol{y}_1, \boldsymbol{y}_2, \ldots, \boldsymbol{y}_k\}$ denotes the set of all measurements available up to time step k.

Besides this optimality property, the Kalman filter allows to incorporate stochastic system models, account for uncertainties in both the system dynamics and measurements, and recursively updates the state estimate in a computationally efficient manner. This makes the Kalman filter well suited for the estimation problem considered in this thesis. In particular, atmospheric turbulence, as captured by the Dryden model (Section 2.3.1), is described as a stochastic process driven by white noise. Therefore, the Dryden turbulence model can be naturally integrated into the Kalman filtering framework to estimate the unknown wind acting on the airship.

The Kalman filter framework treats the system dynamics and measurement process as stochastic rather than deterministic. Within this framework, the state $\boldsymbol{x}_k$ and measurement $\boldsymbol{y}_k$ at each time step k are modeled as Gaussian random vectors rather than deterministic quantities. A Gaussian random vector is fully characterized by its mean and covariance matrix. The mean represents the expected value of the random vector and is defined as

$$\bar{\boldsymbol{x}} = \mathbb{E}[\boldsymbol{x}], \tag{2.51}$$

while the covariance matrix describes its deviation from the mean and is defined as

$$\boldsymbol{P}_{xx} = \mathbb{E}\left[(\boldsymbol{x} - \bar{\boldsymbol{x}})(\boldsymbol{x} - \bar{\boldsymbol{x}})^T\right]. \tag{2.52}$$

The covariance matrix serves as a measure of the uncertainty associated with a random vector. The diagonal entries of $\boldsymbol{P}_{xx}$ represent the variances of the individual components and quantify their spread around the mean, where larger variances indicate greater uncertainty. The off-diagonal entries describe the correlation between the components and indicate how the uncertainty of one component is related to the uncertainty of another.

To describe the evolution of the state over time and how the measurements relate to the system state, the Kalman filter uses a discrete-time probabilistic state-space model. To account for modeling inaccuracies and measurement uncertainties, as well as the stochastic wind, the process and measurement models are augmented by additive random vectors representing process noise and measurement noise, respectively.

$$\boldsymbol{x}_k = \boldsymbol{A}_{k-1}\boldsymbol{x}_{k-1} + \boldsymbol{B}_{k-1}\boldsymbol{u}_{k-1} + \boldsymbol{q}_{k-1}, \tag{2.53}$$

$$\boldsymbol{y}_k = \boldsymbol{C}_k\boldsymbol{x}_k + \boldsymbol{r}_k, \tag{2.54}$$

where $\boldsymbol{x}_k \in \mathbb{R}^n$ denotes the system state, $\boldsymbol{y}_k \in \mathbb{R}^m$ the measurement, $\boldsymbol{u}_k \in \mathbb{R}^p$ the known deterministic input, $\boldsymbol{A}_{k-1} \in \mathbb{R}^{n\times n}$ the state transition matrix, $\boldsymbol{B}_{k-1} \in \mathbb{R}^{n\times p}$ the input matrix, and $\boldsymbol{C}_k \in \mathbb{R}^{m\times n}$ the measurement matrix. The system matrices may be time-varying. The random vectors $\boldsymbol{q}_{k-1} \in \mathbb{R}^n$ and $\boldsymbol{r}_k \in \mathbb{R}^m$ represent the process and measurement noise, respectively. Both are modeled as zero-mean, white Gaussian noise processes with covariance matrices $\boldsymbol{Q}_{k-1} \in \mathbb{R}^{n\times n}$ and $\boldsymbol{R}_k \in \mathbb{R}^{m\times m}$,

$$\mathbb{E}[\boldsymbol{q}_k] = 0, \quad \mathbb{E}[\boldsymbol{q}_k\boldsymbol{q}_k^T] = \boldsymbol{Q}_k \qquad \text{and} \qquad \mathbb{E}[\boldsymbol{r}_k] = 0, \quad \mathbb{E}[\boldsymbol{r}_k\boldsymbol{r}_k^T] = \boldsymbol{R}_k \qquad \forall k. \tag{2.55}$$

As a consequence of this formulation, the state and the measurement are themselves random vectors rather than deterministic quantities. Therefore, the initial condition is not a fixed value but a initial distribution,

$$\boldsymbol{x}_0 \sim \mathcal{N}(\hat{\boldsymbol{x}}_0, \boldsymbol{P}_0), \tag{2.56}$$

with initial state estimate $\hat{\boldsymbol{x}}_0 \in \mathbb{R}^n$ and covariance $\boldsymbol{P}_0 \in \mathbb{R}^{n\times n}$ assumed given.

The state estimate at any time step is defined as the mean of the state's probability distribution.

$$\hat{\boldsymbol{x}}_k = \mathbb{E}[\boldsymbol{x}_k] = \bar{\boldsymbol{x}}_k \tag{2.57}$$

Because the initial state $\boldsymbol{x}_0$ is Gaussian, the process and measurement noise are Gaussian, and both the prediction and update steps introduced below act linearly on Gaussian random vectors, the conditional distribution of the state remains Gaussian at every time step, both before and after a measurement is incorporated,

$$\boldsymbol{x}_k \mid \boldsymbol{y}_{1:k-1} \sim \mathcal{N}(\hat{\boldsymbol{x}}_k^-, \boldsymbol{P}_k^-), \qquad \boldsymbol{x}_k \mid \boldsymbol{y}_{1:k} \sim \mathcal{N}(\hat{\boldsymbol{x}}_k, \boldsymbol{P}_k). \tag{2.58}$$

Here the superscript $(\cdot)^-$ denotes *a priori* quantities, computed before incorporating the measurement at time step k, while quantities without superscript denote *a posteriori* quantities, computed after incorporating the measurement. In both cases, the state estimate is given by the conditional mean of the state, whereas the associated covariance quantifies the remaining uncertainty of that estimate. For notational simplicity, the explicit conditioning on the available measurements is omitted in the remainder of this section.

Starting from the initial estimate $\hat{\boldsymbol{x}}_0$, the Kalman filter recursively computes the state estimate at each time step k by alternating between two steps: a prediction step and an update step. In the prediction step, the previous a posteriori state estimate $\hat{\boldsymbol{x}}_{k-1}$ and the system model are used to propagate the estimate and its associated uncertainty forward in time, yielding the a priori estimate $\hat{\boldsymbol{x}}_k^-$ and covariance $\boldsymbol{P}_k^-$. Additionally, the predicted measurement corresponding to $\hat{\boldsymbol{x}}_k^-$ is calculated.

In the subsequent update step, the latest measurement is incorporated to correct the a priori estimate $\hat{\boldsymbol{x}}_k^-$. The correction is based on the so-called innovation, i.e., the difference between the actual and predicted measurement, and is weighted according to the relative uncertainties of the prediction and the measurement. The resulting a posteriori estimate $\hat{\boldsymbol{x}}_k$, incorporates all available information up to time step k. The corresponding equations are introduced in the following [**SÃďrkkÃď2013KF**].

Prediction

By taking the expectation of Eq. (2.53) and using $\mathbb{E}[\boldsymbol{q}_{k-1}] = 0$, the a priori estimate computes to

$$\hat{\boldsymbol{x}}_k^- = \mathbb{E}[\boldsymbol{x}_k] = \boldsymbol{A}_{k-1}\hat{\boldsymbol{x}}_{k-1} + \boldsymbol{B}_{k-1}\boldsymbol{u}_{k-1}, \tag{2.59}$$

which can be computed for all k recursively given the input $\boldsymbol{u}_k$ and initial estimate $\hat{\boldsymbol{x}}_0$.

The a priori covariance at time step k is defined as

$$\boldsymbol{P}_k^- = \mathbb{E}\left[\left(\boldsymbol{x}_k - \hat{\boldsymbol{x}}_k^-\right)\left(\boldsymbol{x}_k - \hat{\boldsymbol{x}}_k^-\right)^T\right]. \tag{2.60}$$

Subtracting Eq. (2.59) from Eq. (2.53), the known input term $\boldsymbol{B}_{k-1}\boldsymbol{u}_{k-1}$ cancels, leaving the propagated estimation error

$$\boldsymbol{x}_k - \hat{\boldsymbol{x}}_k^- = \boldsymbol{A}_{k-1}\left(\boldsymbol{x}_{k-1} - \hat{\boldsymbol{x}}_{k-1}\right) + \boldsymbol{q}_{k-1}. \tag{2.61}$$

Substituting Eq. (2.61) into the covariance definition (2.60) yields

$$\begin{aligned}
\boldsymbol{P}_k^- &= \mathbb{E}\Big[\big(\boldsymbol{A}_{k-1}(\boldsymbol{x}_{k-1} - \hat{\boldsymbol{x}}_{k-1}) + \boldsymbol{q}_{k-1}\big)\big(\boldsymbol{A}_{k-1}(\boldsymbol{x}_{k-1} - \hat{\boldsymbol{x}}_{k-1}) + \boldsymbol{q}_{k-1}\big)^T\Big] \\
&= \underbrace{\mathbb{E}\Big[\boldsymbol{A}_{k-1}(\boldsymbol{x}_{k-1} - \hat{\boldsymbol{x}}_{k-1})(\boldsymbol{x}_{k-1} - \hat{\boldsymbol{x}}_{k-1})^T\boldsymbol{A}_{k-1}^T\Big]}_{\boldsymbol{A}_{k-1}\boldsymbol{P}_{k-1}\boldsymbol{A}_{k-1}^T} + \underbrace{\mathbb{E}\Big[\boldsymbol{q}_{k-1}\boldsymbol{q}_{k-1}^T\Big]}_{\boldsymbol{Q}_{k-1}} \\
&\quad + 2\underbrace{\mathbb{E}\Big[\boldsymbol{A}_{k-1}(\boldsymbol{x}_{k-1} - \hat{\boldsymbol{x}}_{k-1})\boldsymbol{q}_{k-1}^T\Big]}_{=\boldsymbol{0}} \\
&= \boldsymbol{A}_{k-1}\boldsymbol{P}_{k-1}\boldsymbol{A}_{k-1}^T + \boldsymbol{Q}_{k-1}
\end{aligned} \tag{2.62}$$

The cross term vanishes because the process noise $\boldsymbol{q}_{k-1}$ is assumed to be uncorrelated with the state estimation error $\boldsymbol{x}_{k-1} - \hat{\boldsymbol{x}}_{k-1}$. Equations (2.59) and (2.62) together describe how the a priori mean and covariance of the state evolve under the linear system model. Accordingly, the predicted output computes to

$$\hat{\boldsymbol{y}}_k^- = \mathbb{E}[\boldsymbol{y}_k] = \boldsymbol{C}_k\hat{\boldsymbol{x}}_k^-. \tag{2.63}$$

This concludes the prediction step.

Update

In the update step, the a priori estimate $\hat{\boldsymbol{x}}_k^-$ obtained from the prediction step is corrected using the latest measurement $\boldsymbol{y}_k$, resulting in the a posteriori estimate $\hat{\boldsymbol{x}}_k$. The correction is expressed as a linear combination of the a priori estimate and the innovation, weighted by the Kalman gain $\boldsymbol{K}_k$,

$$\hat{\boldsymbol{x}}_k = \hat{\boldsymbol{x}}_k^- + \boldsymbol{K}_k \tilde{\boldsymbol{y}}_k, \tag{2.64}$$

where the innovation $\tilde{\boldsymbol{y}}_k$ is defined as the difference between the actual and predicted measurement,

$$\tilde{\boldsymbol{y}}_k = \boldsymbol{y}_k - \hat{\boldsymbol{y}}_k^- = \boldsymbol{C}_k(\boldsymbol{x}_k - \hat{\boldsymbol{x}}_k^-) + \boldsymbol{r}_k. \tag{2.65}$$

The Kalman gain $\boldsymbol{K}_k \in \mathbb{R}^{n\times m}$ is chosen such that the a posteriori estimate $\hat{\boldsymbol{x}}_k$ in Eq. (2.64) minimizes the trace of the a posteriori covariance $\boldsymbol{P}_k$, consistent with the MMSE criterion in Eq. (2.50). The resulting optimal gain is given by [**SÃďrkkÃď2013KF**]

$$\boldsymbol{K}_k = \boldsymbol{P}_k^{xy} \boldsymbol{S}_k^{-1}, \tag{2.66}$$

where $\boldsymbol{S}_k \in \mathbb{R}^{m\times m}$ denotes the innovation covariance, which quantifies the uncertainty associated with the innovation. Assuming that the measurement noise $\boldsymbol{r}_k$ is uncorrelated with the a priori estimation error $\boldsymbol{x}_k - \hat{\boldsymbol{x}}_k^-$, the innovation covariance can be computed using Eqs. (2.54) and (2.63) as

$$\boldsymbol{S}_k = \mathbb{E}[\tilde{\boldsymbol{y}}_k \tilde{\boldsymbol{y}}_k^T] = \underbrace{\mathbb{E}\big[\boldsymbol{C}_k(\boldsymbol{x}_k - \hat{\boldsymbol{x}}_k^-)(\boldsymbol{x}_k - \hat{\boldsymbol{x}}_k^-)^T \boldsymbol{C}_k^T\big]}_{\boldsymbol{C}_k \boldsymbol{P}_k^- \boldsymbol{C}_k^T} + \underbrace{\mathbb{E}[\boldsymbol{r}_k \boldsymbol{r}_k^T]}_{\boldsymbol{R}_k} = \boldsymbol{C}_k \boldsymbol{P}_k^- \boldsymbol{C}_k^T + \boldsymbol{R}_k, \tag{2.67}$$

and $\boldsymbol{P}_k^{xy} \in \mathbb{R}^{n\times m}$ denotes the cross-covariance between the a priori state estimation error and the innovation, given by

$$\boldsymbol{P}_k^{xy} = \mathbb{E}\big[(\boldsymbol{x}_k - \hat{\boldsymbol{x}}_k^-)\tilde{\boldsymbol{y}}_k^T\big] = \mathbb{E}\big[(\boldsymbol{x}_k - \hat{\boldsymbol{x}}_k^-)(\boldsymbol{x}_k - \hat{\boldsymbol{x}}_k^-)^T\big]\boldsymbol{C}_k^T = \boldsymbol{P}_k^- \boldsymbol{C}_k^T. \tag{2.68}$$

The a posteriori covariance follows by substituting Eqs. (2.65) and (2.64) into the definition of the estimation error,

$$\boldsymbol{x}_k - \hat{\boldsymbol{x}}_k = (\boldsymbol{x}_k - \hat{\boldsymbol{x}}_k^-) - \boldsymbol{K}_k \tilde{\boldsymbol{y}}_k = (\boldsymbol{I} - \boldsymbol{K}_k \boldsymbol{C}_k)(\boldsymbol{x}_k - \hat{\boldsymbol{x}}_k^-) - \boldsymbol{K}_k \boldsymbol{r}_k, \tag{2.69}$$

so that, following the same expansion as in Eq. (2.67),

$$\boldsymbol{P}_k = (\boldsymbol{I} - \boldsymbol{K}_k \boldsymbol{C}_k)\boldsymbol{P}_k^-(\boldsymbol{I} - \boldsymbol{K}_k \boldsymbol{C}_k)^T + \boldsymbol{K}_k \boldsymbol{R}_k \boldsymbol{K}_k^T. \tag{2.70}$$

Substituting the optimal gain from Eq. (2.66) simplifies this expression to

$$\boldsymbol{P}_k = \boldsymbol{P}_k^- - \boldsymbol{K}_k \boldsymbol{S}_k \boldsymbol{K}_k^T = (\boldsymbol{I} - \boldsymbol{K}_k \boldsymbol{C}_k)\boldsymbol{P}_k^-. \tag{2.71}$$

This concludes the update step. Together, the prediction and update steps in Eqs. (2.59), (2.62), (2.64), and (2.71) form the recursive Kalman filter algorithm, alternately propagating the a priori estimate forward in time and correcting it into an a posteriori estimate as new measurements become available.

The recursive state estimate can be summarized compactly as

$$\hat{\boldsymbol{x}}_k = \underbrace{\boldsymbol{A}_{k-1}\hat{\boldsymbol{x}}_{k-1} + \boldsymbol{B}_{k-1}\boldsymbol{u}_{k-1}}_{\text{Prediction}} + \underbrace{\boldsymbol{K}_k\left(\boldsymbol{y}_k - \hat{\boldsymbol{y}}_k\right)}_{\text{Update}}, \tag{2.72}$$

where the first term predicts the state based on the system model, while the second term corrects this prediction using the innovation weighted by the Kalman gain.

This predictor-corrector structure is common to many state estimation methods, including deterministic observers such as the Luenberger observer [**Luenberger1964**, **Luenberger1966**]. The distinguishing feature of the Kalman filter is that it not only estimates the system state but also recursively propagates the associated estimation uncertainty through the error covariance matrix.

The presented equations describe the Kalman filter formulation for linear systems. In many real-world applications, however, the system dynamics and measurement models are inherently nonlinear. This is also the case for the airship model considered in this work, whose nonlinear equations of motion require a nonlinear state estimation approach.

Several methods have been proposed to extend the Kalman filter framework to nonlinear systems, including the Extended Kalman Filter (EKF) and the Unscented Kalman Filter (UKF) [**Maybeck1979**, **Simon2006**, **Julier1997**]. Both methods retain the recursive prediction-update structure of the classical Kalman filter while employing different strategies to handle nonlinearities. These two approaches are introduced in the following.

2.4.1. Extended Kalman Filter

The nonlinear discrete-time probabilistic state-space model is given by

$$\boldsymbol{x}_k = \boldsymbol{f}(\boldsymbol{x}_{k-1}, \boldsymbol{u}_{k-1}) + \boldsymbol{q}_{k-1}, \tag{2.73}$$

$$\boldsymbol{y}_k = \boldsymbol{h}(\boldsymbol{x}_k) + \boldsymbol{r}_k, \tag{2.74}$$

where $\boldsymbol{f}$ and $\boldsymbol{h}$ denote the, possibly time-varying, nonlinear state transition and measurement functions, respectively, replacing the matrices $\boldsymbol{A}_{k-1}$, $\boldsymbol{B}_{k-1}$, and $\boldsymbol{C}_k$ of the linear model in Eq. (2.53). The process noise $\boldsymbol{q}_{k-1}$ and measurement noise $\boldsymbol{r}_k$ retain the same statistical properties as in the linear case.

The Extended Kalman Filter (EKF) extends the classical Kalman filter to nonlinear systems by linearizing the nonlinear system around the latest available state estimate [**Simon2006**]. This results in a locally linear system representation, allowing the recursive prediction and update structure of the Kalman filter to be applied. Specifically, the nonlinear state transition function $\boldsymbol{f}$ is linearized around the latest *a posteriori* state estimate $\hat{\boldsymbol{x}}_{k-1}$ during the prediction step, while the nonlinear measurement function $\boldsymbol{h}$ is linearized around the latest *a priori* state estimate $\hat{\boldsymbol{x}}_k^-$ during the update step. The resulting first-order approximations are given by [**SÃďrkkÃď2013KF**]

$$\boldsymbol{x}_k \approx \boldsymbol{f}(\hat{\boldsymbol{x}}_{k-1}, \boldsymbol{u}_{k-1}) + \boldsymbol{F}_{k-1}(\boldsymbol{x}_{k-1} - \hat{\boldsymbol{x}}_{k-1}) + \boldsymbol{q}_{k-1}, \tag{2.75}$$

$$\boldsymbol{y}_k \approx \boldsymbol{h}(\hat{\boldsymbol{x}}_k^-) + \boldsymbol{H}_k(\boldsymbol{x}_k - \hat{\boldsymbol{x}}_k^-) + \boldsymbol{r}_k, \tag{2.76}$$

where $\boldsymbol{F}_{k-1} \in \mathbb{R}^{n\times n}$ denotes the Jacobian of $\boldsymbol{f}$ with respect to the state, evaluated at $(\hat{\boldsymbol{x}}_{k-1}, \boldsymbol{u}_{k-1})$, and $\boldsymbol{H}_k \in \mathbb{R}^{m\times n}$ denotes the Jacobian of $\boldsymbol{h}$ with respect to the state, evaluated at $\hat{\boldsymbol{x}}_k^-$,

$$\boldsymbol{F}_{k-1} = \left.\frac{\partial \boldsymbol{f}}{\partial \boldsymbol{x}}\right|_{\hat{\boldsymbol{x}}_{k-1}, \boldsymbol{u}_{k-1}} \qquad \text{and} \qquad \boldsymbol{H}_k = \left.\frac{\partial \boldsymbol{h}}{\partial \boldsymbol{x}}\right|_{\hat{\boldsymbol{x}}_k^-} \tag{2.77}$$

Prediction

Analogously to the classical Kalman filter, the *a priori* state estimate is obtained by propagating the previous *a posteriori* estimate through the system model. Taking the expectation of the linearized state equation in Eq. (2.75) yields

$$\begin{aligned}\hat{\boldsymbol{x}}_k^- &= \mathbb{E}[\boldsymbol{x}_k] = \boldsymbol{f}(\hat{\boldsymbol{x}}_{k-1}, \boldsymbol{u}_{k-1}) + \boldsymbol{F}_{k-1}\underbrace{\mathbb{E}[\boldsymbol{x}_{k-1} - \hat{\boldsymbol{x}}_{k-1}]}_{=\boldsymbol{0}} + \underbrace{\mathbb{E}[\boldsymbol{q}_{k-1}]}_{=\boldsymbol{0}} \\ &= \boldsymbol{f}(\hat{\boldsymbol{x}}_{k-1}, \boldsymbol{u}_{k-1}),\end{aligned} \tag{2.78}$$

whereas the second term computes to zero because the expected value of the state estimation error computes to zero: $\mathbb{E}[\boldsymbol{x}_{k-1} - \hat{\boldsymbol{x}}_{k-1}] = \hat{\boldsymbol{x}}_{k-1} - \hat{\boldsymbol{x}}_{k-1} = \boldsymbol{0}$.

The a priori covariance follows from the state estimation error

$$\boldsymbol{x}_k - \hat{\boldsymbol{x}}_k^- = \boldsymbol{F}_{k-1}(\boldsymbol{x}_{k-1} - \hat{\boldsymbol{x}}_{k-1}) + \boldsymbol{q}_{k-1}. \tag{2.79}$$

Equation (2.79) has the same structure as in the linear case in Eq. (2.61), with $\boldsymbol{A}_{k-1}$ replaced by the Jacobian $\boldsymbol{F}_{k-1}$. Applying the linear covariance propagation result from Eq. (2.62) to this locally linearized system gives the a priori covariance

$$\boldsymbol{P}_k^- = \boldsymbol{F}_{k-1}\boldsymbol{P}_{k-1}\boldsymbol{F}_{k-1}^T + \boldsymbol{Q}_{k-1}. \tag{2.80}$$

The predicted measurement follows analogously by evaluating the nonlinear measurement function at the a priori estimate,

$$\hat{\boldsymbol{y}}_k^- = \boldsymbol{h}(\hat{\boldsymbol{x}}_k^-). \tag{2.81}$$

This concludes the prediction step.

Update

As in the linear case, the a posteriori estimate is obtained by correcting the a priori estimate with the innovation,

$$\hat{\boldsymbol{x}}_k = \hat{\boldsymbol{x}}_k^- + \boldsymbol{K}_k\tilde{\boldsymbol{y}}_k, \tag{2.82}$$

where the innovation is given by

$$\tilde{\boldsymbol{y}}_k = \boldsymbol{y}_k - \hat{\boldsymbol{y}}_k^- = \boldsymbol{H}_k(\boldsymbol{x}_k - \hat{\boldsymbol{x}}_k^-) + \boldsymbol{r}_k. \tag{2.83}$$

Equation (2.83) has the same structure as in the linear case, with $\boldsymbol{C}_k$ replaced by the Jacobian $\boldsymbol{H}_k$. Applying the linear update equations from Eqs. (2.66)–(2.71) to this locally linearized system yields the innovation covariance, Kalman gain, and a posteriori covariance,

$$\boldsymbol{S}_k = \boldsymbol{H}_k \boldsymbol{P}_k^- \boldsymbol{H}_k^T + \boldsymbol{R}_k, \tag{2.84}$$

$$\boldsymbol{K}_k = \boldsymbol{P}_k^- \boldsymbol{H}_k^T \boldsymbol{S}_k^{-1}, \tag{2.85}$$

$$\boldsymbol{P}_k = (\boldsymbol{I} - \boldsymbol{K}_k \boldsymbol{H}_k)\boldsymbol{P}_k^-. \tag{2.86}$$

This concludes the update step.

The EKF approximates the nonlinear state transition and measurement functions by neglecting higher-order terms of the Taylor expansion and propagating only the first-order statistics of the state distribution. Consequently, the accuracy of the filter depends on the degree of nonlinearity of the system and the validity of the local linear approximation [**Simon2006**]. In contrast to the linear Kalman filter, the Jacobians $\boldsymbol{F}_{k-1}$ and $\boldsymbol{H}_k$ depend on the current state estimate and must therefore be recomputed at every time step, increasing the computational cost. Although this approach sacrifices the exact optimality guarantees of the linear Kalman filter, the EKF remains one of the most widely used nonlinear estimation methods due to its conceptual simplicity and comparatively low computational overhead relative to alternative nonlinear estimation techniques.

2.4.2. Unscented Kalman Filter

The unscented Kalman filter (UKF) avoids the local linearization employed by the EKF by propagating a deterministically chosen set of sample points, referred to as *sigma points*, through the nonlinear process and measurement functions $\boldsymbol{f}$ and $\boldsymbol{h}$ [**Julier2004UKF**]. The transformed sigma points are then used to reconstruct the mean and covariance of the propagated probability distribution using the so-called *unscented transform*. Since the nonlinear functions are evaluated directly rather than approximated by a first-order Taylor expansion, the unscented transform captures the mean and covariance of the transformed distribution with at least second-order accuracy for arbitrary nonlinear functions and third-order accuracy for Gaussian distributions, while avoiding the computation of Jacobians [**Julier2004UKF**].

Sigma Points and the Unscented Transform

The unscented transform is a method for approximating the mean and covariance of a random variable after undergoing a nonlinear transformation $\boldsymbol{y} = \boldsymbol{g}(\boldsymbol{x})$. The probability distribution of the n-dimensional Gaussian random vector $\boldsymbol{x}$ is represented by a set of $2n+1$ sigma points $\boldsymbol{\mathcal{X}}_i \in \mathbb{R}^n$, which are deterministically generated from the current mean $\bar{\boldsymbol{x}}$ and covariance $\boldsymbol{P}$ as [**Julier1997UKF**]

$$\begin{aligned}
\boldsymbol{\mathcal{X}}_0 &= \bar{\boldsymbol{x}}, \\
\boldsymbol{\mathcal{X}}_i &= \bar{\boldsymbol{x}} + \sqrt{(n+\lambda)}\left[\sqrt{\boldsymbol{P}}\right]_i, \qquad i = 1,\dots,n, \\
\boldsymbol{\mathcal{X}}_{i+n} &= \bar{\boldsymbol{x}} - \sqrt{(n+\lambda)}\left[\sqrt{\boldsymbol{P}}\right]_i, \qquad i = 1,\dots,n,
\end{aligned} \tag{2.87}$$

where $[\sqrt{\boldsymbol{P}}]_i$ denotes the i-th column of a matrix square root satisfying $\sqrt{\boldsymbol{P}}\sqrt{\boldsymbol{P}}^T = \boldsymbol{P}$ and λ is a scaling parameter defined as

$$\lambda = \alpha^2(n+\kappa) - n. \tag{2.88}$$

The parameters α, β, and κ determine the spread of the sigma points around the mean. A common default choice is $\kappa = 0$ with α chosen small and positive to keep the sigma points close to $\bar{\boldsymbol{x}}$. For Gaussian random vectors $\beta = 2$ is optimal [**Wan2001UKF**].

The sigma points $\boldsymbol{\mathcal{X}}_i$ are then propagated through the nonlinear function

$$\boldsymbol{\mathcal{Y}}_i = \boldsymbol{g}(\boldsymbol{\mathcal{X}}_i), \qquad i = 0, \dots, 2n \tag{2.89}$$

to obtain the set of transformed sigma points $\boldsymbol{\mathcal{Y}}_i$. The mean and covariance of the transformed distribution $\boldsymbol{y}$ are reconstructed as the weighted sample mean and covariance of the propagated sigma points

$$\bar{\boldsymbol{y}} = \sum_{i=0}^{2n} W_i^m \boldsymbol{\mathcal{Y}}_i, \tag{2.90}$$

$$\boldsymbol{P}_{yy} = \sum_{i=0}^{2n} W_i^c \left(\boldsymbol{\mathcal{Y}}_i - \bar{\boldsymbol{y}}\right)\left(\boldsymbol{\mathcal{Y}}_i - \bar{\boldsymbol{y}}\right)^T, \tag{2.91}$$

with weights for mean W_i^m and covariance W_i^c defined as [**Wan2001UKF**]

$$\begin{aligned}
W_0^m &= \frac{\lambda}{n+\lambda}, \\
W_0^c &= \frac{\lambda}{n+\lambda} + (1 - \alpha^2 + \beta), \\
W_i^m = W_i^c &= \frac{1}{2(n+\lambda)}, \qquad i = 1, \dots, 2n.
\end{aligned} \tag{2.92}$$

Since the weights depend only on the state dimension n and the scaling parameters α, β, and κ, they remain constant throughout the filtering process and can therefore be computed once prior to filter execution. In the UKF algorithm, the unscented transform is applied separately during the prediction and update steps by propagating the sigma points through the nonlinear state transition function $\boldsymbol{f}$ and the nonlinear measurement function $\boldsymbol{h}$, respectively.

Prediction

A set of sigma points $\boldsymbol{\mathcal{X}}_{k-1,i}$ is generated according to Eq. (2.87) from the previous a posteriori estimate $\hat{\boldsymbol{x}}_{k-1}$ and covariance $\boldsymbol{P}_{k-1}$. The sigma points are then propagated individually through the nonlinear state transition function,

$$\boldsymbol{\mathcal{X}}_{k,i}^- = \boldsymbol{f}(\boldsymbol{\mathcal{X}}_{k-1,i}, \boldsymbol{u}_{k-1}), \qquad i = 0, \dots, 2n. \tag{2.93}$$

Applying the unscented transform in Eqs. (2.90) and (2.91) to the propagated sigma points yields the a priori estimate and covariance [**Wan2001UKF**],

$$\hat{\boldsymbol{x}}_k^- = \sum_{i=0}^{2n} W_i^m \boldsymbol{\mathcal{X}}_{k,i}^-, \tag{2.94}$$

$$\boldsymbol{P}_k^- = \sum_{i=0}^{2n} W_i^c \left(\boldsymbol{\mathcal{X}}_{k,i}^- - \hat{\boldsymbol{x}}_k^-\right)\left(\boldsymbol{\mathcal{X}}_{k,i}^- - \hat{\boldsymbol{x}}_k^-\right)^T + \boldsymbol{Q}_{k-1}, \tag{2.95}$$

where the process noise covariance $\boldsymbol{Q}_{k-1}$ is added directly, consistent with the additive noise formulation in Eq. (2.73).

To incorporate the computed a priori estimate into the output prediction, a new set of sigma points $\boldsymbol{\mathcal{X}}_{k,i}$ is generated according to Eq. (2.87) from the a priori estimate $\hat{\boldsymbol{x}}_k^-$ and covariance $\boldsymbol{P}_k^-$. The predicted measurement is obtained by propagating the redrawn sigma points through the nonlinear measurement function and applying the unscented transform (2.90)

$$\boldsymbol{\mathcal{Y}}_{k,i} = \boldsymbol{h}(\boldsymbol{\mathcal{X}}_{k,i}), \qquad i = 0, \ldots, 2n, \tag{2.96}$$

$$\hat{\boldsymbol{y}}_k^- = \sum_{i=0}^{2n} W_i^m \boldsymbol{\mathcal{Y}}_{k,i}. \tag{2.97}$$

This concludes the prediction step.

Update

As in the linear Kalman filter and the EKF, the a posteriori estimate is obtained by correcting the a priori estimate with the innovation,

$$\hat{\boldsymbol{x}}_k = \hat{\boldsymbol{x}}_k^- + \boldsymbol{K}_k \tilde{\boldsymbol{y}}_k, \tag{2.98}$$

The innovation covariance and the cross-covariance between the a priori state and the predicted measurement are reconstructed from the propagated sigma points using the unscented transform,

$$\boldsymbol{S}_k = \sum_{i=0}^{2n} W_i^c \left(\boldsymbol{\mathcal{Y}}_{k,i} - \hat{\boldsymbol{y}}_k^-\right)\left(\boldsymbol{\mathcal{Y}}_{k,i} - \hat{\boldsymbol{y}}_k^-\right)^T + \boldsymbol{R}_k, \tag{2.99}$$

$$\boldsymbol{P}_k^{xy} = \sum_{i=0}^{2n} W_i^c \left(\boldsymbol{\mathcal{X}}_{k,i}^- - \hat{\boldsymbol{x}}_k^-\right)\left(\boldsymbol{\mathcal{Y}}_{k,i} - \hat{\boldsymbol{y}}_k^-\right)^T, \tag{2.100}$$

where the measurement noise covariance $\boldsymbol{R}_k$ is added directly to $\boldsymbol{S}_k$, analogous to $\boldsymbol{Q}_{k-1}$ in Eq. (2.95). As in the linear Kalman filter, the Kalman gain and a posteriori covariance follow from Eqs. (2.66) and (2.71),

$$\boldsymbol{K}_k = \boldsymbol{P}_k^{xy} \boldsymbol{S}_k^{-1}, \tag{2.101}$$

$$\boldsymbol{P}_k = \boldsymbol{P}_k^- - \boldsymbol{K}_k \boldsymbol{S}_k \boldsymbol{K}_k^T. \tag{2.102}$$

This concludes the update step.

The UKF approximates the propagation of the state distribution by deterministically sampling a set of sigma points and propagating them through the nonlinear process and measurement functions. Consequently, it captures the effect of nonlinear transformations on the mean and covariance more accurately than the EKF, while avoiding the first-order linearization error associated with Taylor series approximations [**Julier2004UKF**]. In contrast to the EKF, no Jacobians of $\boldsymbol{f}$ or $\boldsymbol{h}$ are required. Instead, the nonlinear functions are evaluated directly at $2n+1$ sigma points, resulting in a computational cost that scales with the state dimension. Jacobian evaluation. Although the UKF still relies on a finite set of sample points and therefore does not provide the exact optimality guarantees of the linear Kalman filter, its improved accuracy relative to the EKF at comparable computational cost has made it a widely used alternative for nonlinear state estimation.

2.5. Disturbance Compensation

Compensation methods aim to improve the disturbance rejection and tracking performance of a control system by reducing the influence of known or estimated disturbances and compensating for modeled system dynamics. Rather than relying solely on feedback to correct errors after they occur, compensation methods utilize additional information, such as system models, disturbance estimates, or reference dynamics, to generate corrective control actions proactively. The effectiveness of such approaches depends strongly on the accuracy of the system model and the available disturbance information. Consequently, compensation methods are typically combined with a feedback controller, which compensates for remaining modeling errors and unmeasured disturbances while ensuring closed-loop stability [**Astrom2010FeedbackSystems**, **Lunze2013Regelungstechnik2**].

The general structure of a compensation-based control approach is depicted in Fig. 2.4.

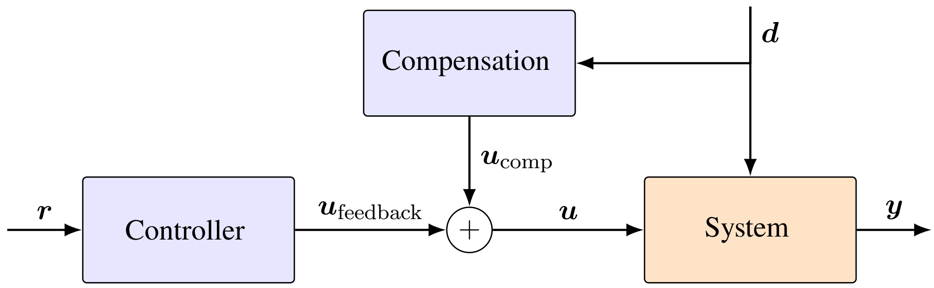


Figure 2.4.: Structure of a compensation-based control approach. Both the controller and the compensation module may additionally utilize available system outputs of the airship. For clarity, the corresponding signal paths are omitted.

2.5.1. Disturbance Feedforward

Disturbance feedforward (DFF) is an explicit model-based disturbance compensation strategy that utilizes measurements or estimates of external disturbances to generate a compensating control input before the disturbance affects the controlled output.

Consider the linear time-invariant (LTI) system

$$\dot{\boldsymbol{x}} = \boldsymbol{A}\boldsymbol{x} + \boldsymbol{B}\boldsymbol{u} + \boldsymbol{E}\boldsymbol{d}, \tag{2.103}$$

where $\boldsymbol{x} \in \mathbb{R}^n$ denotes the system state, $\boldsymbol{u} \in \mathbb{R}^p$ the control input, and $\boldsymbol{d} \in \mathbb{R}^q$ a measurable or estimated disturbance acting on the system through the disturbance input matrix $\boldsymbol{E} \in \mathbb{R}^{n \times q}$.

Now the feedback control law is augmented with a disturbance feedforward compensation term

$$\boldsymbol{u} = \boldsymbol{u}_{\text{feedback}} + \boldsymbol{u}_{\text{comp}}, \tag{2.104}$$

where $\boldsymbol{u}_{\text{feedback}}$ denotes the nominal feedback control input and $\boldsymbol{u}_{\text{comp}}$ the additional compensation input designed to counteract the disturbance influence. The compensation input can be selected as [**Lunze2013Regelungstechnik2**]

$$\boldsymbol{u}_{\text{comp}} = -\boldsymbol{B}^{\dagger}\boldsymbol{E}\boldsymbol{d}, \tag{2.105}$$

where $\boldsymbol{B}^{\dagger}$ denotes the Moore-Penrose pseudoinverse of the input matrix. For a square and invertible input matrix, the pseudoinverse reduces to the ordinary inverse, i.e., $\boldsymbol{B}^{\dagger} = \boldsymbol{B}^{-1}$. In this case, the compensated closed-loop dynamics become

$$\dot{\boldsymbol{x}} = \boldsymbol{A}\boldsymbol{x} + \boldsymbol{B}\boldsymbol{u}_{\text{feedback}}, \tag{2.106}$$

which are independent of the disturbance.

The same principle can be extended to nonlinear systems. Consider the nonlinear input affine dynamics [**Isidori1995NonlinearControlSystems**]

$$\dot{\boldsymbol{x}} = \boldsymbol{f}(\boldsymbol{x}) + \boldsymbol{g}(\boldsymbol{x})\boldsymbol{u} + \boldsymbol{g}_d(\boldsymbol{x})\boldsymbol{d}, \tag{2.107}$$

where $\boldsymbol{f} : \mathbb{R}^n \to \mathbb{R}^n$, $\boldsymbol{g} : \mathbb{R}^n \to \mathbb{R}^{n \times m}$, and $\boldsymbol{g}_d : \mathbb{R}^n \to \mathbb{R}^{n \times q}$ denote the nonlinear system dynamics, the input distribution matrix, and the disturbance distribution matrix, respectively. As for the linear case, we augment the control input with an compensation feedforward term

$$\boldsymbol{u}_{\text{comp}} = -\boldsymbol{g}^{\dagger}(\boldsymbol{x})\boldsymbol{g}_d(\boldsymbol{x})\boldsymbol{d}, \tag{2.108}$$

For an invertible input matrix $\boldsymbol{g}(\boldsymbol{x})$, this results in the compensated dynamics

$$\dot{\boldsymbol{x}} = \boldsymbol{f}(\boldsymbol{x}) + \boldsymbol{g}(\boldsymbol{x})\boldsymbol{u}_{\text{feedback}}, \tag{2.109}$$

which are independent of the disturbance.

2.5.2. Nonlinear Dynamic Inversion

Nonlinear dynamic inversion (NDI) is a model-based control strategy that applies the principles of feedback linearization. NDI exploits an accurate nonlinear model of the system to algebraically cancel its known nonlinear dynamics and transform the nonlinear system into an approximately linear input-output behavior [**Stevens2015AircraftControl**, **Slotine1991AppliedNC**]. In contrast to disturbance feedforward, NDI does not only compensate the effect of external disturbances but incorporates them into the nonlinear system model and computes the control input required to compensate the complete known system dynamics.

The nonlinear dynamics are described by

$$\dot{\boldsymbol{x}} = \boldsymbol{f}(\boldsymbol{x}, \boldsymbol{d}) + \boldsymbol{g}(\boldsymbol{x})\boldsymbol{u}, \tag{2.110}$$

where $\boldsymbol{f} : \mathbb{R}^n \times \mathbb{R}^q \rightarrow \mathbb{R}^n$ represents the known nonlinear system dynamics including the influence of the disturbance and $\boldsymbol{g} : \mathbb{R}^n \rightarrow \mathbb{R}^{n \times m}$ denotes the input distribution matrix.

The control input is augmented as

$$\boldsymbol{u} = \boldsymbol{u}_{\text{feedback}} + \boldsymbol{u}_{\text{comp}}, \tag{2.111}$$

where the feedback term is responsible for defining the desired closed-loop behavior, whereas the compensation term is designed to cancel the known nonlinear system dynamics and disturbance effects. The compensation term is therefore selected as:

$$\boldsymbol{u}_{\text{comp}} = -\boldsymbol{g}^{\dagger}(\boldsymbol{x})\boldsymbol{f}(\boldsymbol{x}, \boldsymbol{d}). \tag{2.112}$$

Substituting the compensation term into the system dynamics (2.110) results in the compensated dynamics assuming that the nonlinear dynamics can be exactly inverted through the input distribution matrix

$$\dot{\boldsymbol{x}} = \boldsymbol{g}(\boldsymbol{x})\boldsymbol{u}_{\text{feedback}}. \tag{2.113}$$

Thus, the nonlinearities and known disturbance effects included in $\boldsymbol{f}(\boldsymbol{x}, \boldsymbol{d})$ are removed from the system description, leaving the feedback controller to determine the remaining closed-loop behavior.

3. model-based wind estimation and compensation

This chapter presents the implementation of the proposed model-based wind estimation and compensation framework.The first part focuses on the implementation of the Kalman filter used for wind estimation, including the required model formulation and discretization. In the second part, the estimated wind velocity is utilized to implement the disturbance compensation methods.

3.1. Kalman Filter

The implementation of the Kalman filter requires a discrete-time state-space model of the airship dynamics that captures the influence of wind. Starting from the nonlinear airship model, the Dryden wind model is incorporated to obtain an augmented state-space representation. Subsequently, the continuous-time system is discretized and the corresponding process noise covariance matrices are derived. Based on this discrete-time model, the Extended Kalman Filter (EKF) and the Unscented Kalman Filter (UKF) are implemented for recursive wind estimation.

3.1.1. State Space Description

A valid and complete state space description is the fundamental prerequisite for implementing a Kalman filter. While the filter algorithm itself follows a well-defined prediction and correction procedure, its performance depends heavily on an accurate mathematical representation of the system dynamics and the available measurements [**SÃďrkka2013KF**]. Consequently, deriving an appropriate state space description is the first step toward implementing the wind estimation and compensation strategy presented in this thesis.

The airship model derived in Section 2.2 describes the translational and rotational dynamics using the generalized velocity vector

$$ {}_B\boldsymbol{\nu} = \begin{bmatrix} {}_B\boldsymbol{v}_B \\ {}_B\boldsymbol{\omega}_B \end{bmatrix}, \tag{3.1} $$

where ${}_B\boldsymbol{v}_B$ and ${}_B\boldsymbol{\omega}_B$ denote the translational and angular velocities expressed in the body-fixed coordinate system. The rigid-body dynamics depend on the generalized forces acting at the COG and have been derived as

$$ \boldsymbol{M}\,{}_B\dot{\boldsymbol{\nu}} = {}_B\boldsymbol{\mathcal{F}}^{\mathrm{COG}}_{\mathrm{grav}} + {}_B\boldsymbol{\mathcal{F}}^{\mathrm{COG}}_{\mathrm{buoy}} + {}_B\boldsymbol{\mathcal{F}}^{\mathrm{COG}}_{\mathrm{aero}} + {}_B\boldsymbol{\mathcal{F}}^{\mathrm{COG}}_{\mathrm{prop}} - {}_B\boldsymbol{\mathcal{F}}^{\mathrm{COG}}_{\mathrm{cor}}. \tag{3.2} $$

Although Eq. (3.2) describes the motion of the airship, it does not represent a complete state space description yet.

First, the generalized forces acting on the airship do not depend on $\boldsymbol{\nu}$ alone. For example, $\boldsymbol{\mathcal{F}}_{\text{grav}}$ and $\boldsymbol{\mathcal{F}}_{\text{buoy}}$ must be transformed into the body-fixed coordinate system and thus depend on the attitude $\boldsymbol{\Theta}$ of the airship. To obtain a state-space model whose state evolution depends only on the current state and inputs, the attitude must be included in the state vector.

ref

Furthermore, the state-space model requires an output equation relating the state to the available onboard sensor measurements. As described in , the airship is equipped with sensors providing measurements of its position and attitude. Therefore, to express the measurements as a function of the state, both position $\boldsymbol{p}$ and attitude $\boldsymbol{\Theta}$ must be included in the state.

Consequently, the state vector is defined as

$$\boldsymbol{x} = \begin{bmatrix} {}_E\boldsymbol{p} \\ \boldsymbol{\Theta} \\ {}_B\boldsymbol{v}_B \\ {}_B\boldsymbol{\omega}_B \end{bmatrix} = \begin{bmatrix} x \\ y \\ z \\ \phi \\ \theta \\ \psi \\ u \\ v \\ w \\ p \\ q \\ r \end{bmatrix}, \tag{3.3}$$

where ${}_E\boldsymbol{p}$ represents the position in the earth-fixed coordinate system, whose kinematics are given by

$${}_E\dot{\boldsymbol{p}} = \begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{z} \end{bmatrix} = \boldsymbol{R}_{EB\,B}\boldsymbol{v}_B, \tag{3.4}$$

and $\boldsymbol{\Theta}$ contains the attitude angles, with respective kinematics [1]

$$\dot{\boldsymbol{\Theta}} = \begin{bmatrix} \dot{\phi} \\ \dot{\theta} \\ \dot{\psi} \end{bmatrix} = \begin{bmatrix} 1 & \sin\phi\tan\theta & \cos\phi\tan\theta \\ 0 & \cos\phi & -\sin\phi \\ 0 & \dfrac{\sin\phi}{\cos\theta} & \dfrac{\cos\phi}{\cos\theta} \end{bmatrix} {}_B\boldsymbol{\omega}_B = \boldsymbol{J}(\boldsymbol{\Theta})\,{}_B\boldsymbol{\omega}_B. \tag{3.5}$$

Combining the rigid-body dynamics in Eq. (3.2) with the kinematic equations for position and attitude yields the nonlinear state-space model

$$\dot{\boldsymbol{x}} = \begin{bmatrix} {}_E\dot{\boldsymbol{p}} \\ \dot{\boldsymbol{\Theta}} \\ {}_B\dot{\boldsymbol{v}}_B \\ {}_B\dot{\boldsymbol{\omega}}_B \end{bmatrix} = \begin{bmatrix} \boldsymbol{R}_{EB}\,{}_B\boldsymbol{v}_B \\ \boldsymbol{J}(\boldsymbol{\Theta})\,{}_B\boldsymbol{\omega}_B \\ \boldsymbol{M}^{-1}\left(\boldsymbol{\mathcal{F}}_{\text{grav}}(\boldsymbol{x}) + \boldsymbol{\mathcal{F}}_{\text{buoy}}(\boldsymbol{x}) + \boldsymbol{\mathcal{F}}_{\text{aero}}(\boldsymbol{x}, \boldsymbol{v}_{\boldsymbol{A}}) + \boldsymbol{\mathcal{F}}_{\text{prop}}(\boldsymbol{u}) - \boldsymbol{\mathcal{F}}_{\text{cor}}(\boldsymbol{x})\right) \end{bmatrix}. \tag{3.6}$$

$$\boldsymbol{y} = \boldsymbol{h}(\boldsymbol{x}) = \boldsymbol{x}. \tag{3.7}$$

However, the aerodynamic forces and moments $\boldsymbol{\mathcal{F}}_{\mathrm{aero}}$ depend explicitly on the aerodynamic velocity, which is influenced by the wind velocity. Therefore, the wind dynamics must be incorporated into the state representation to obtain a complete model suitable for wind estimation. The incorporation of the wind model and the resulting augmented state space description are presented in Section 3.1.2.

The state space model derived above represents a deterministic description of the airship dynamics. In practice, however, both the system model and the sensor measurements are subject to uncertainty. Modeling inaccuracies arise primarily from the aerodynamic and propulsion models due to parameter uncertainties, simplifications, and unmodeled effects, while sensor measurements are affected by measurement noise. To account for these uncertainties, the system and measurement equations are augmented with stochastic process and measurement noise, respectively.

Since the airship dynamics are formulated on a force and moment level, the process noise is introduced at the same level. This preserves the physical structure of the equations of motion while directly representing uncertainties in the modeled aerodynamic and propulsion forces and moments. The generalized force and moment vector is therefore modeled as

$$\boldsymbol{\mathcal{F}} = \boldsymbol{\mathcal{F}}_{\mathrm{model}} + \boldsymbol{\zeta} \qquad \text{with} \qquad \boldsymbol{\zeta} \sim \mathcal{N}\left(\mathbf{0}, \boldsymbol{\Sigma}_{\zeta}\right), \tag{3.8}$$

where $\boldsymbol{\mathcal{F}}_{\mathrm{model}}$ denotes the nominal generalized force and moment vector obtained from the airship model, and $\boldsymbol{\zeta}$ represents the modeling uncertainty associated with the aerodynamic and propulsion components. To reflect the assumption that modeling errors increase with the magnitude of the acting forces and moments, the standard deviation is chosen proportional to the corresponding aerodynamic and propulsion generalized forces,

$$\boldsymbol{\sigma}_{\zeta} = \alpha_{\mathrm{aero}} \left|\boldsymbol{\mathcal{F}}_{\mathrm{aero}}\right| + \alpha_{\mathrm{prop}} \left|\boldsymbol{\mathcal{F}}_{\mathrm{prop}}\right|, \tag{3.9}$$

where $|\cdot|$ denotes the element wise absolute value and α_{aero} and α_{prop} are tuning parameters that determine the relative uncertainty associated with the aerodynamic and propulsion models. In this work, both parameters are chosen as $\alpha_{\mathrm{aero}} = \alpha_{\mathrm{prop}} = 0.1$ corresponding to a standard deviation of each uncertain force and moment component equal to 10% of its magnitude.

The covariance matrix used for the process noise is then obtained as

$$\boldsymbol{\Sigma}_{\zeta} = \mathrm{diag}\left(\boldsymbol{\sigma}_{\zeta}^{2}\right). \tag{3.10}$$

Similarly, the measurement noise is modeled as additive zero-mean Gaussian noise acting on the measurement equation

$$\boldsymbol{y} = \boldsymbol{h}(\boldsymbol{x}) + \boldsymbol{\xi} \qquad \text{with} \qquad \boldsymbol{\xi} \sim \mathcal{N}\left(\mathbf{0}, \boldsymbol{\Sigma}_{\xi}\right). \tag{3.11}$$

Assuming uncorrelated noise across the individual sensor channels, the measurement noise covariance follows directly from the selected standard deviations,

$$\boldsymbol{\Sigma}_{\xi} = \mathrm{diag}\left(\boldsymbol{\sigma}_{\xi}^{2}\right), \tag{3.12}$$

where $\boldsymbol{\sigma}_{\xi}$ contains conservative estimates for the standard deviations of the individual measured quantities. The chosen measurement noise standard deviations are summarized in Appendix **??**.

3.1.2. Including the Wind Model

As stated in Section 2.3, the wind is modeled as the sum of constant wind and turbulence described by the Dryden turbulence model,

$$ {}_E\boldsymbol{v}_W = {}_E\boldsymbol{v}_{W,\text{const}} + {}_E\boldsymbol{v}_{W,\text{turb}}. \tag{3.13} $$

To incorporate the wind model into the airship state space description, a state space realization of the wind dynamics is required. The dynamics of the constant wind component are simply given by

$$ {}_E\dot{\boldsymbol{v}}_{\text{const}} = \boldsymbol{0}. \tag{3.14} $$

Hence, the dynamic part of the wind model is entirely described by the Dryden turbulence model. The turbulent wind component is generated by passing white Gaussian noise through the Dryden shaping filters introduced in Section 2.3.1. Consequently, state space realizations of the transfer functions in Eqs. (2.43), (2.44), and (2.45) are required. Using the controllable canonical form [**Lunze2012Regelungstechnik1**] and introducing the white Gaussian noise $\eta_i \sim \mathcal{N}(0,1)$, the first-order transfer function for the longitudinal wind component yields

$$ \dot{x}_u = \underbrace{-\frac{V}{L_u}}_{=:\boldsymbol{A}_u} x_u + \eta_u, \tag{3.15} $$

$$ {}_E u_W = \underbrace{\sigma_u \sqrt{\frac{2V}{\pi L_u}}}_{=:\boldsymbol{C}_u} x_u. \tag{3.16} $$

The lateral and vertical shaping filters share the same state space structure. As both are of second order their state space representation is of dimension two. For the generic wind component $i \in \{v, w\}$, the realization is given by

$$ \dot{\boldsymbol{x}}_i = \underbrace{\begin{bmatrix} 0 & 1 \\ -\frac{V^2}{L_i^2} & -\frac{2V}{L_i} \end{bmatrix}}_{=:\boldsymbol{A}_i} \boldsymbol{x}_i + \begin{bmatrix} 0 \\ 1 \end{bmatrix} \eta_i, \tag{3.17} $$

$$ {}_E i_W = \underbrace{\begin{bmatrix} \sigma_i \frac{V}{L_i} \sqrt{\frac{V}{\pi L_i}} & \sigma_i \sqrt{\frac{3V}{\pi L_i}} \end{bmatrix}}_{=:\boldsymbol{C}_i} \boldsymbol{x}_i. \tag{3.18} $$

As defined in Section 2.3.1, the scale lengths L_i depend on the flight altitude $h = -z$. The reference wind speed is assumed constant with $W_{20} = 10\,\mathrm{m/s}$, allowing the turbulence intensities σ_i to be precomputed. This assumption is justified in Section 5.1.1. Furthermore, since $V = \|{}_B\boldsymbol{v}_B\|_2$, the system matrices $\boldsymbol{A}_i$ and $\boldsymbol{C}_i$ depend on the airship state.

Combining the state space realization for longitudinal, lateral and vertical wind direction yields the probabilistic state space description

$$\dot{\boldsymbol{x}}_W = \underbrace{\begin{bmatrix} \boldsymbol{A}_u & \boldsymbol{0} & \boldsymbol{0} \\ \boldsymbol{0} & \boldsymbol{A}_v & \boldsymbol{0} \\ \boldsymbol{0} & \boldsymbol{0} & \boldsymbol{A}_w \end{bmatrix}}_{=:\boldsymbol{A}_W(\boldsymbol{x})} \boldsymbol{x}_W + \underbrace{\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}}_{=:\boldsymbol{B}_W} \boldsymbol{\eta}, \tag{3.19}$$

$$_E\boldsymbol{v}_W = \underbrace{\begin{bmatrix} \boldsymbol{C}_u & \boldsymbol{0} & \boldsymbol{0} \\ \boldsymbol{0} & \boldsymbol{C}_v & \boldsymbol{0} \\ \boldsymbol{0} & \boldsymbol{0} & \boldsymbol{C}_w \end{bmatrix}}_{=:\boldsymbol{C}_W(\boldsymbol{x})} \boldsymbol{x}_W, \tag{3.20}$$

where $\boldsymbol{A}_W(\boldsymbol{x}) \in \mathbb{R}^{5\times5}$ and $\boldsymbol{C}_W(\boldsymbol{x}) \in \mathbb{R}^{3\times5}$ depend on the state $\boldsymbol{x}$. The wind model can now be appended to the airship state space model, yielding the augmented state vector

$$\tilde{\boldsymbol{x}} = \begin{bmatrix} \boldsymbol{x} \\ \boldsymbol{x}_W \end{bmatrix}. \tag{3.21}$$

Although the complete wind model consists of a constant and a turbulent component, only the Dryden filter states are appended to the system state as they describe the wind dynamics. Consequently, the augmented model contains only the five Dryden states, representing the internal states of the Dryden shaping filters. They can be used to describe the wind velocity according to Eq.(3.20). As a result the aerodynamic velocity is given by

$$_B\boldsymbol{v}_A = {}_B\boldsymbol{v}_B - {}_B\boldsymbol{v}_W = {}_B\boldsymbol{v}_B - \boldsymbol{R}_{BE}\boldsymbol{C}_W(\boldsymbol{x})\boldsymbol{x}_W, \tag{3.22}$$

which allows all aerodynamic forces and moments to be expressed as functions of the augmented state. This completes the nonlinear state space description of the airship, which is now given in the structure

$$\dot{\tilde{\boldsymbol{x}}} = \begin{bmatrix} \dot{\boldsymbol{x}} \\ \dot{\boldsymbol{x}}_W \end{bmatrix} = \begin{bmatrix} \boldsymbol{f}(\tilde{\boldsymbol{x}}, \boldsymbol{u}) \\ \boldsymbol{A}_W \boldsymbol{x}_W \end{bmatrix} + \begin{bmatrix} \boldsymbol{M}^{-1}\boldsymbol{\zeta} \\ \boldsymbol{B}_W \boldsymbol{\eta} \end{bmatrix}, \tag{3.23}$$

$$\boldsymbol{y} = \boldsymbol{h}(\tilde{\boldsymbol{x}}) + \boldsymbol{\xi}. \tag{3.24}$$

3.1.3. Airspeed Sensor

measurement noise from sensor specifications Pitot tube only measurements if $u > 2m/s$. To account for this we implement a time varying measurement noise, setting $R_{pitottube} \rightarrow \infty$ if airspeed in x-direction is < 2. To understand why this works lets look at the computation of the kalman gain. R -> infinity simply means that we fully trust our dynamics and basically do not use the measurement.

3.1.4. Discretization

The Kalman filter framework requires a discrete-time state transition function and measurement function. Therefore, the continuous-time nonlinear state space model derived in the previous sections is transformed into the discrete form

$$\tilde{\boldsymbol{x}}_k = \boldsymbol{f}_d(\tilde{\boldsymbol{x}}_{k-1}, \boldsymbol{u}_{k-1}) + \boldsymbol{q}_{k-1}, \tag{3.25}$$

$$\boldsymbol{y}_k = \boldsymbol{h}_d(\tilde{\boldsymbol{x}}_k) + \boldsymbol{r}_k. \tag{3.26}$$

where the discrete process and measurement noise are assumed to be zero-mean Gaussian stochastic processes,

$$\boldsymbol{q}_k \sim \mathcal{N}(\boldsymbol{0}, \boldsymbol{Q}_k), \qquad \boldsymbol{r}_k \sim \mathcal{N}(\boldsymbol{0}, \boldsymbol{R}_k). \tag{3.27}$$

The process noise covariance combines the uncertainty of the nonlinear airship model and the discretized Dryden wind model,

$$\boldsymbol{Q}_k = \mathrm{blkdiag}\left(\boldsymbol{Q}_{k,\mathrm{model}}, \boldsymbol{Q}_{k,\mathrm{wind}}\right). \tag{3.28}$$

Since the augmented system consists of a nonlinear airship model and a linear wind model, both subsystems are discretized separately.

An exact analytical discretization of the nonlinear airship dynamics is generally not available. Therefore, the state transition is approximated using the forward Euler method,

$$\boldsymbol{x}_k = \boldsymbol{x}_{k-1} + T_s \boldsymbol{f}(\tilde{\boldsymbol{x}}_{k-1}, \boldsymbol{u}_{k-1}) + \boldsymbol{q}_{k-1,\mathrm{model}}, \tag{3.29}$$

where $T_s = 0.01\,\mathrm{s}$ denotes the sampling time. The forward Euler method exhibits a global error of order $\mathcal{O}(T_s)$. Although higher-order integration methods, such as the classical fourth-order Runge–Kutta method, provide greater numerical accuracy, they require multiple evaluations of the nonlinear dynamics per sampling step. Given the small sampling time and the presence of model uncertainties represented by the process noise, the additional computational effort is not justified for the present application.

The continuous-time force and moment uncertainty introduced in Section 3.1.1 is transformed into an equivalent discrete-time covariance. Following [**BarShalom2001Discretization**] and approximating $e^{\boldsymbol{A}T_s} \approx \boldsymbol{I}$, the discrete covariance is obtained as

$$\boldsymbol{Q}_{k-1,\mathrm{model}} = \boldsymbol{M}^{-1}\boldsymbol{\Sigma}_\zeta(\tilde{\boldsymbol{x}}_{k-1})\left(\boldsymbol{M}^{-1}\right)^T. \tag{3.30}$$

For fixed airship states, the Dryden shaping filters constitute a linear state-space model. Consequently, the wind dynamics can be discretized exactly using the matrix exponential,

$$\boldsymbol{x}_{W,k} = \boldsymbol{A}_{W,k-1}\boldsymbol{x}_{W,k-1} + \boldsymbol{q}_{k-1,\mathrm{wind}}, \tag{3.31}$$

with the discrete state transition matrix [**Astrom1997Discretization**]

$$\boldsymbol{A}_{W,k-1} = e^{\boldsymbol{A}_W T_s}\Big|_{\boldsymbol{x}_{k-1}}. \tag{3.32}$$

The corresponding discrete process noise covariance is given by [**BarShalom2001Discretization**]

$$\boldsymbol{Q}_{k-1,\mathrm{wind}} = \int_0^{T_s} e^{\boldsymbol{A}_W\tau}\boldsymbol{B}_W\boldsymbol{\Sigma}_\eta\boldsymbol{B}_W^T e^{\boldsymbol{A}_W^T\tau}\,d\tau\Big|_{\boldsymbol{x}_{k-1}}, \tag{3.33}$$

where $\boldsymbol{\Sigma}_\eta = \boldsymbol{I}_{3\times 3}$ since the driving noise processes are independent standard Gaussian white noise processes. The integral is evaluated using the van Loan method [**VanLoan1978**].

Finally, the discretized airship dynamics and the discretized Dryden wind model are combined to obtain the complete discrete-time state transition function

$$\boldsymbol{f}_d(\tilde{\boldsymbol{x}}_{k-1}, \boldsymbol{u}_{k-1}) = \begin{bmatrix} \boldsymbol{x}_{k-1} + T_s\, \boldsymbol{f}(\tilde{\boldsymbol{x}}_{k-1}, \boldsymbol{u}_{k-1}) \\ \boldsymbol{A}_{W,k-1}\, \boldsymbol{x}_{W,k-1} \end{bmatrix}, \tag{3.34}$$

with process noise covariance $\boldsymbol{Q}_{k-1}$ as given above. Since the measurement model introduced in Section 3.1.1 is a static, memoryless function of the state, its discretization is trivial, $\boldsymbol{h}_d(\tilde{\boldsymbol{x}}_k) = \boldsymbol{h}(\tilde{\boldsymbol{x}}_k)$, and no separate treatment is required.

3.1.5. Extended Kalman Filter Implementation

The implementation of the EKF directly follows the prediction and update equations introduced in Section 2.4.1. Rather than reiterating the standard algorithm, this section discusses the implementation choices made in this work. In particular, it addresses the computation of the Jacobian matrices and a numerically robust formulation of the covariance update.

Computation of Jacobians

The EKF requires the Jacobians $\boldsymbol{F}_k$ and $\boldsymbol{H}_k$ of the discrete-time state transition function $\boldsymbol{f}_d$ and measurement function $\boldsymbol{h}_d$ with respect to the augmented state $\tilde{\boldsymbol{x}}$. Both Jacobians are evaluated at each sampling instant using the current state estimate. An analytical derivation of the state transition Jacobian $\boldsymbol{F}_k$ is not practical for the considered system. The nonlinear aerodynamic force and moment models, together with their coupling to the wind states through the relative airspeed, result in lengthy expressions that are computationally inefficient to derive symbolically and evaluate online in the Simulink environment. Instead, $\boldsymbol{F}_k$ is approximated numerically using a forward finite-difference scheme,

$$[\boldsymbol{F}_k]_i \approx \frac{\boldsymbol{f}_d\Big(\hat{\tilde{\boldsymbol{x}}}_{k-1} + \varepsilon \boldsymbol{e}_i, \boldsymbol{u}_{k-1}\Big) - \boldsymbol{f}_d\Big(\hat{\tilde{\boldsymbol{x}}}_{k-1}, \boldsymbol{u}_{k-1}\Big)}{\varepsilon}, \tag{3.35}$$

where $\boldsymbol{e}_i$ denotes the i-th unit vector and $[\boldsymbol{F}_k]_i$ the i-th column of $\boldsymbol{F}_k$. A constant perturbation size of $\varepsilon = 10^{-6}$ is used. A forward finite-difference approximation was chosen due to its lower computational cost compared with central differencing while still providing sufficient accuracy for the present application.

In contrast, the measurement Jacobian $\boldsymbol{H}_k$ is computed analytically from the measurement model. The comparatively simple structure of the measurement model allows for a straightforward analytical derivation, making a numerical approximation unnecessary.

Robust Covariance Update

The conventional EKF covariance update given in (2.86),

$$\boldsymbol{P}_k = (\boldsymbol{I} - \boldsymbol{K}_k \boldsymbol{H}_k)\, \boldsymbol{P}_k^-, \tag{3.36}$$

can be sensitive to numerical errors. Inaccuracies in the Kalman gain $\boldsymbol{K}_k$, resulting for example from the numerical approximation of the state transition Jacobian, finite-precision arithmetic, or an ill-conditioned innovation covariance matrix, may cause the covariance matrix $\boldsymbol{P}_k$ to lose symmetry or positive semidefiniteness during simulation. If left unaddressed, these numerical effects can accumulate over time and may lead to filter divergence.

To improve numerical robustness, the covariance update is implemented using the Joseph form [**Simon2006KF**],

$$\boldsymbol{P}_k = (\boldsymbol{I} - \boldsymbol{K}_k\boldsymbol{H}_k)\,\boldsymbol{P}_k^-\,(\boldsymbol{I} - \boldsymbol{K}_k\boldsymbol{H}_k)^T + \boldsymbol{K}_k\boldsymbol{R}_k\boldsymbol{K}_k^T, \tag{3.37}$$

which better preserves the symmetry and positive semi-definiteness of the covariance matrix in finite-precision arithmetic. This improved numerical stability comes at the cost of additional matrix multiplications compared with the conventional covariance update.

3.1.6. Unscented Kalman Filter Implementation

The implementation of the UKF directly follows the prediction and update equations introduced in Section 2.4.2. Therefore, this section focuses on the implementation choices specific to the augmented airship–wind model, in particular the selection of the sigma point scaling parameters and the computation of the matrix square root.

Sigma Point Parameters

The unscented transform requires the selection of three scaling parameters: α, β, and κ. The parameter α determines the spread of the sigma points around the estimated mean, β incorporates prior knowledge about the distribution and is optimal at $\beta = 2$ for Gaussian distributions, while κ serves as an additional scaling parameter.

Following the scaled unscented transform formulation in [**Julier2004Unscented**],κ is set to $\kappa = 0$, which represents a commonly used practical choice. The parameter β is set to $\beta = 2$ as we consider Gaussian state distributions.

The choice of α has a significant influence on the sigma point distribution and therefore on the filter performance. Instead of selecting this parameter manually, its value is determined through a Bayesian optimization procedure [**Snoek2012Bayesian**] using the wind estimation performance as the optimization objective. Based on this optimization, $\alpha = 0.01$ is selected for the UKF implementation.

Matrix Square Root Computation

The unscented transform requires the computation of the matrix square root of the state covariance matrix in order to generate the sigma points. In the present implementation, this operation is performed using a Cholesky decomposition, which provides an efficient and numerically stable factorization for positive definite covariance matrices,

$$\boldsymbol{P}_k = \boldsymbol{L}_k\boldsymbol{L}_k^T. \tag{3.38}$$

However, due to numerical errors introduced during the covariance propagation and update steps, the covariance matrix may no longer be exactly symmetric or positive semidefinite in finite-precision arithmetic. Since the Cholesky decomposition requires a positive definite matrix, these small numerical deviations can cause the decomposition to fail.

To improve numerical robustness, the covariance matrix is therefore regularized prior to the Cholesky decomposition. First, symmetry is enforced by averaging the covariance matrix with its transpose

$$\boldsymbol{P}_k \leftarrow \frac{1}{2}\left(\boldsymbol{P}_k + \boldsymbol{P}_k^T\right). \tag{3.39}$$

Subsequently, an eigenvalue decomposition is performed, and negative or numerically insignificant eigenvalues are replaced by a small positive tolerance. The covariance matrix is then reconstructed from the modified eigenvalues and symmetrized again before applying the Cholesky decomposition. This procedure ensures that the covariance matrix remains suitable for the sigma point generation while preserving the original covariance structure as closely as possible.

3.1.7. Filter Initialization

The filter is initialized with a zero state estimate and a small diagonal initial covariance matrix,

$$\hat{\tilde{\boldsymbol{x}}}_0 = \boldsymbol{0}, \qquad \boldsymbol{P}_0 = 10^{-3}\boldsymbol{I}, \tag{3.40}$$

where $\boldsymbol{I}$ denotes the identity matrix of the augmented state dimension. The zero initial state estimate eliminates the need for prior knowledge of the true initial condition. A small initial covariance is selected to limit large corrections and thus ensure a smooth convergence behavior during the first filter iterations. The influence of the initial estimate and covariance is mainly limited to the transient phase, during which the filter converges from the initial estimate toward the actual state trajectory. After convergence, the filter behavior is primarily determined by the process noise covariance $\boldsymbol{Q}_{k-1}$, the measurement noise covariance $\boldsymbol{R}_k$, and the innovation sequence. Therefore, the specific choice of $\hat{\tilde{\boldsymbol{x}}}_0$ and $\boldsymbol{P}_0$ has no significant influence on the long-term estimation performance.

The EKF and UKF estimate the complete augmented state vector. The wind estimate is subsequently obtained from the estimated wind states through the wind model output equation (3.20)

$$\boxed{{}_E\hat{\boldsymbol{v}}_W = \boldsymbol{C}_W(\hat{\boldsymbol{x}})\hat{\boldsymbol{x}}_W.} \tag{3.41}$$

3.2. Compensation

While wind is modeled as an additional system state in the Kalman Filter implementation, it can also be considered as an external disturbance acting on the airship. Therefore, the wind estimation provided by the Kalman Filter can be utilized to compensate for the effect of wind disturbances. Two compensation approaches are implemented and evaluated: a nonlinear Disturbance Feedforward (DFF) controller and a Nonlinear Dynamic Inversion (NDI) controller. Both approaches can be interpreted as an augmentation to the existing PID controller, adding an additional compensation term

$$\boldsymbol{u} = \boldsymbol{u}_{\text{PID}} + \boldsymbol{u}_{\text{comp}}, \tag{3.42}$$

where $\boldsymbol{u}_{\text{PID}}$ is mainly responsible for setpoint tracking and stabilization, while $\boldsymbol{u}_{\text{comp}}$ uses model knowledge of the airship to counteract the wind-induced disturbances in case of DFF, or compensates for all acting forces and moments in case of the NDI.

To minimize the delay between wind disturbance estimation and compensation, the estimated disturbance is compensated directly at the force and moment level. Based on the estimated wind and the current airship state measurement, the required compensation force and moment vector

$$\boldsymbol{\mathcal{F}}_{\text{comp}} = \begin{bmatrix} \boldsymbol{F}_{\text{comp}} \\ \boldsymbol{M}_{\text{comp}} \end{bmatrix} \in \mathbb{R}^6 \tag{3.43}$$

is calculated. The computation of $\boldsymbol{\mathcal{F}}_{\text{comp}}$ depends on the compensation approach and is derived in the sections 3.2.1 and 3.2.2.

The total desired force and moment vector is obtained by combining the force and moment vector generated by the PID controller, $\boldsymbol{\mathcal{F}}_{\text{PID}}$, with the compensation vector, $\boldsymbol{\mathcal{F}}_{\text{comp}}$:

$$\boldsymbol{\mathcal{F}}_{\text{des}} = \boldsymbol{\mathcal{F}}_{\text{PID}} + \boldsymbol{\mathcal{F}}_{\text{comp}}. \tag{3.44}$$

This force and moment vector is subsequently converted into individual motor commands by the control allocation algorithm. Performing the compensation directly at the force and moment level enables computationally efficient execution, as the mapping from forces and moments to motor commands $\boldsymbol{u}$ is performed only once per time step. At the same time, the underlying concept introduced in Eq. (3.42) is preserved, allowing disturbances to be compensated with a delay of only one control cycle.

$\boldsymbol{\mathcal{F}}_{\text{des}}$ denotes the desired force and moment generated by the propulsion system. However, due to the individual actuator limits of the motors, the achieved force and moment, $\boldsymbol{\mathcal{F}}_{\text{prop}}$, may differ from the desired value.

The resulting control structure, including the compensation strategy, is illustrated in Figure 3.1.

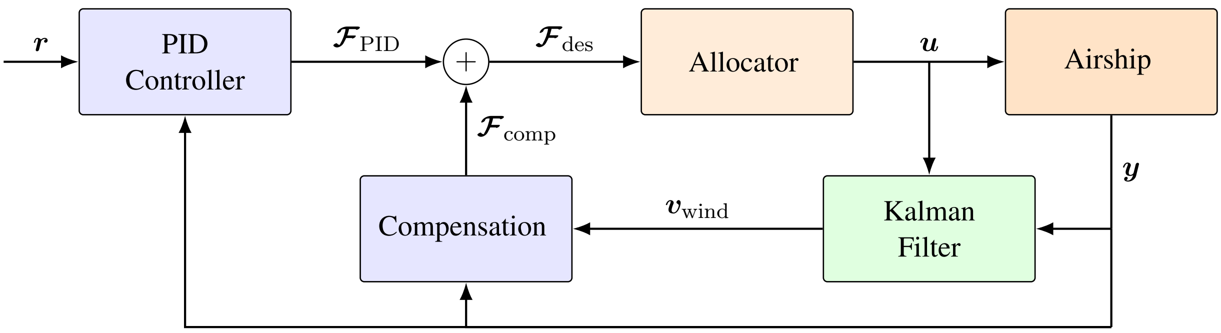


Figure 3.1.: Block diagram of the control structure, showing the control structure with combined PID and compensation.

The original PID controller, without compensation, was designed with a strong emphasis on robustness against modeling uncertainties and wind disturbances. Since the compensation approaches explicitly account for a significant portion of wind-induced disturbances, the PID controller gains can be retuned to prioritize tracking performance. In Section 5.2, the effects of the proposed compensation methods on the control performance are evaluated. To investigate the influence of the controller tuning, multiple sets of PID gains are considered in combination with the different compensation approaches.

In the following sections, the two compensation approaches are presented in detail, and the corresponding compensation vector $\boldsymbol{\mathcal{F}}_{\text{comp}}$ is derived. It is assumed that the airship operates under nominal flight conditions, such that the desired force and moment generated by the control system can be realized by the propulsion system without actuator saturation. Therefore

$$\boldsymbol{\mathcal{F}}_{\text{prop}} = \boldsymbol{\mathcal{F}}_{\text{des}} = \boldsymbol{\mathcal{F}}_{\text{PID}} + \boldsymbol{\mathcal{F}}_{\text{comp}}. \tag{3.45}$$

In addition, we assume that exact state measurements from the Sensor Fusion of the airship are available.

3.2.1. Disturbance Feedforward

The Disturbance Feedforward approach compensates only for the additional forces and moments induced by wind disturbances. Therefore, the compensation vector $\boldsymbol{\mathcal{F}}_{\text{comp}}$ is designed to counteract the change in aerodynamic forces and moments caused by the prevailing wind. To determine the wind-induced contribution, the equations of motion introduced in Section 2.2 are considered:

$$\boldsymbol{M}\dot{\boldsymbol{\nu}} = \boldsymbol{\mathcal{F}}_{\text{grav}}(\boldsymbol{x}) + \boldsymbol{\mathcal{F}}_{\text{buoy}}(\boldsymbol{x}) + \boldsymbol{\mathcal{F}}_{\text{aero}}(\boldsymbol{x}, \boldsymbol{v_A}) + \boldsymbol{\mathcal{F}}_{\text{prop}}(\boldsymbol{u}) - \boldsymbol{\mathcal{F}}_{\text{cor}}(\boldsymbol{x}). \tag{3.46}$$

The wind affects the system dynamics only through the aerodynamic generalized forces, as these depend on the aerodynamic velocity

$$\boldsymbol{v}_A = \boldsymbol{v}_B - \boldsymbol{v}_W. \tag{3.47}$$

Consequently, the wind-induced forces and moments can be determined as the difference between the generalized aerodynamic forces in the presence of wind and the corresponding forces in still air. In the latter case, the aerodynamic velocity is equal to the body velocity, i.e., $\boldsymbol{v}_A = \boldsymbol{v}_B$:

$$\Delta\boldsymbol{\mathcal{F}}_{\text{wind}} = \boldsymbol{\mathcal{F}}_{\text{aero}}(\boldsymbol{x}, \boldsymbol{v}_A) - \boldsymbol{\mathcal{F}}_{\text{aero}}(\boldsymbol{x}, \boldsymbol{v}_B). \tag{3.48}$$

Substituting Eqs. (3.48) and (3.45) into the equations of motion (3.46) yields

$$\boldsymbol{M}\dot{\boldsymbol{\nu}} = \boldsymbol{\mathcal{F}}_{\text{grav}}(\boldsymbol{x}) + \boldsymbol{\mathcal{F}}_{\text{buoy}}(\boldsymbol{x}) + \boldsymbol{\mathcal{F}}_{\text{aero}}(\boldsymbol{x}, \boldsymbol{v}_B) - \boldsymbol{\mathcal{F}}_{\text{cor}}(\boldsymbol{x}) + \Delta\boldsymbol{\mathcal{F}}_{\text{wind}} + \boldsymbol{\mathcal{F}}_{\text{PID}} + \boldsymbol{\mathcal{F}}_{\text{comp}}. \tag{3.49}$$

By considering $\Delta\boldsymbol{\mathcal{F}}_{\text{wind}}$ as the disturbance and $\boldsymbol{\mathcal{F}}_{\text{comp}}$ as the compensation input, the system can be interpreted in the nonlinear input-affine form introduced in Eq. (2.107). Since the compensation is directly applied at the generalized force level, the input matrices $\boldsymbol{g}(\boldsymbol{x})$ and $\boldsymbol{g}_d(\boldsymbol{x})$ reduce to identity mappings. The ideal compensation is therefore given by

$$\boldsymbol{\mathcal{F}}_{\text{comp}} = -\Delta\boldsymbol{\mathcal{F}}_{\text{wind}}. \tag{3.50}$$

This results in the compensated dynamics

$$\boldsymbol{M}\dot{\boldsymbol{\nu}} = \boldsymbol{\mathcal{F}}_{\text{grav}}(\boldsymbol{x}) + \boldsymbol{\mathcal{F}}_{\text{buoy}}(\boldsymbol{x}) + \boldsymbol{\mathcal{F}}_{\text{aero}}(\boldsymbol{x}, \boldsymbol{v}_B) - \boldsymbol{\mathcal{F}}_{\text{cor}}(\boldsymbol{x}) + \boldsymbol{\mathcal{F}}_{\text{PID}}, \tag{3.51}$$

which, assuming perfect knowledge of the model and exact estimation of the wind, are independent of the wind-induced forces and moments.

In practice, exact disturbance compensation cannot be achieved due to two main sources of error. The first results from the fact that only an estimate of the wind velocity, $\hat{\boldsymbol{v}}_W$, is available from the Kalman filter. Consequently, the wind-induced forces and moments must be computed using the estimated aerodynamic velocity

$$\hat{\boldsymbol{v}}_A = \boldsymbol{v}_B - \hat{\boldsymbol{v}}_W. \tag{3.52}$$

The second source of error originates from the aerodynamic model $\boldsymbol{\mathcal{F}}_{\text{aero}}$ itself, which is an approximation of the true aerodynamic forces and moments and does not take all physical effects into account. These two error sources directly affect the computation of $\Delta\boldsymbol{\mathcal{F}}_{\text{wind}}$. Therefore, the implemented compensation is given by

$$\boxed{\boldsymbol{\mathcal{F}}_{\text{comp}} = -\Delta\widehat{\boldsymbol{\mathcal{F}}}_{\text{wind}} = -\boldsymbol{\mathcal{F}}_{\text{aero}}(\boldsymbol{x}, \hat{\boldsymbol{v}}_A) + \boldsymbol{\mathcal{F}}_{\text{aero}}(\boldsymbol{x}, \boldsymbol{v}_B),} \tag{3.53}$$

rather than the ideal compensation $-\Delta\boldsymbol{\mathcal{F}}_{\text{wind}}$ derived above.

3.2.2. Nonlinear Dynamic Inversion

In contrast to the Disturbance Feedforward approach, which compensates only for the wind-induced forces and moments, the Nonlinear Dynamic Inversion approach compensates for the complete nonlinear system dynamics. Rather than isolating the disturbance contribution, NDI simultaneously cancels gravitational, buoyant, aerodynamic, and Coriolis forces to approximately transform the nonlinear system into an linear input-output behavior.

Starting again from the equations of motion introduced in Section 2.2,

$$\boldsymbol{M}\dot{\boldsymbol{\nu}} = \boldsymbol{\mathcal{F}}_{\text{grav}}(\boldsymbol{x}) + \boldsymbol{\mathcal{F}}_{\text{buoy}}(\boldsymbol{x}) + \boldsymbol{\mathcal{F}}_{\text{aero}}(\boldsymbol{x}, \boldsymbol{v}_A) + \boldsymbol{\mathcal{F}}_{\text{prop}}(\boldsymbol{u}) - \boldsymbol{\mathcal{F}}_{\text{cor}}(\boldsymbol{x}), \tag{3.54}$$

and replacing the generalized propulsion force by the sum of the PID and compensation terms yields

$$\boldsymbol{M}\dot{\boldsymbol{\nu}} = \boldsymbol{\mathcal{F}}_{\text{grav}}(\boldsymbol{x}) + \boldsymbol{\mathcal{F}}_{\text{buoy}}(\boldsymbol{x}) + \boldsymbol{\mathcal{F}}_{\text{aero}}(\boldsymbol{x}, \boldsymbol{v}_A) - \boldsymbol{\mathcal{F}}_{\text{cor}}(\boldsymbol{x}) + \boldsymbol{\mathcal{F}}_{\text{PID}} + \boldsymbol{\mathcal{F}}_{\text{comp}}. \tag{3.55}$$

This has the same nonlinear input-affine structure as Eq. (2.110), with $\boldsymbol{g}(\boldsymbol{x})$ again reducing to an identity mapping. In contrast to the Disturbance Feedforward approach, the disturbance does not need to be isolated explicitly, but is treated as part of the nonlinear dynamics $\boldsymbol{f}(\boldsymbol{x}, \boldsymbol{d})$. The resulting ideal NDI compensation is therefore given by

$$\boxed{\boldsymbol{\mathcal{F}}_{\text{comp}} = -\Big(\boldsymbol{\mathcal{F}}_{\text{grav}}(\boldsymbol{x}) + \boldsymbol{\mathcal{F}}_{\text{buoy}}(\boldsymbol{x}) + \boldsymbol{\mathcal{F}}_{\text{aero}}(\boldsymbol{x}, \boldsymbol{v}_A) - \boldsymbol{\mathcal{F}}_{\text{cor}}(\boldsymbol{x})\Big).} \tag{3.56}$$

Substituting Eq. (3.56) into Eq. (3.55) exactly cancels the nonlinear dynamics term, leaving

$$\boldsymbol{M}\dot{\boldsymbol{\nu}} = \boldsymbol{\mathcal{F}}_{\text{PID}}. \tag{3.57}$$

Under the assumption of perfect cancellation, Eq. (3.57) represents an exact feedback linearization of the airship dynamics, resulting in a linear closed-loop system. Consequently, the PID controller no longer needs to compensate for the underlying nonlinear dynamics or wind-induced disturbances.

As with the Disturbance Feedforward approach, perfect cancellation cannot be achieved in practice due to wind estimation errors and inaccuracies in the underlying models. First, evaluating $\boldsymbol{\mathcal{F}}_{\mathrm{aero}}(\boldsymbol{x}, \boldsymbol{v}_A)$ requires the aerodynamic velocity $\boldsymbol{v}_A$, whereas only an estimate $\hat{\boldsymbol{v}}_A$ is available obtained according to Eq. (3.52). Second, the models of the gravitational, buoyant, aerodynamic, and Coriolis forces are themselves approximations and do not capture all physical effects acting on the airship.

In contrast to the Disturbance Feedforward approach, which compensates only the wind-induced aerodynamic contribution, NDI relies on accurate models of the complete nonlinear dynamics. Consequently, NDI is considerably more sensitive to modeling errors and generally requires greater control authority, as the complete nonlinear force vector is cancelled at each time step rather than only the wind-induced component.

4. data-driven wind estimation and compensation

5. Evaluation and results

To evaluate the performance of the proposed wind estimation algorithm and the wind compensation strategies, a simulation-based test framework is established. For the wind estimation task, the performance of the Extended Kalman Filter (EKF) and Unscented Kalman Filter (UKF) is compared under identical conditions. For wind compensation, Disturbance Feedforward (DFF) and Nonlinear Dynamic Inversion (NDI) are evaluated with respect to their impact on tracking performance and compared to the baseline PID controller.

To ensure a representative evaluation under different operating conditions, six flight scenarios are considered. These flight scenarios reflect the main real-world operational scenarios.

- Forward flight: slow ($2\,\mathrm{m/s}$) and fast ($4\ \mathrm{m/s}$)
- Sideward flight ($1\ \mathrm{m/s}$)
- Pure rotation around the yaw axis ($5^\circ/\mathrm{s}$)
- Curved flight: forward flight (($1\ \mathrm{m/s}$) combined with a yaw rotation ($3.6^\circ/\mathrm{s}$)
- Hover

Since the performance of the wind estimation and compensation methods depends strongly on the wind conditions, multiple wind scenarios are considered for each flight scenario. These include different constant wind velocities and varying turbulence intensities generated by varying the Dryden model parameter W_{20}, as introduced in Section 2.3.

The root mean square error (RMSE) is used as the primary performance metric throughout this evaluation. It provides a quantitative measure of the deviation between a reference signal and its corresponding estimate or response. For a signal x with N samples, the RMSE is defined as [**Chai2014RMSE**]

$$\mathrm{RMSE}(x) = \sqrt{\frac{1}{N}\sum_{k=1}^{N}(x_k - \hat{x}_k)^2}, \tag{5.1}$$

where x_k denotes the reference value and $\hat{x}_k$ the corresponding estimated or simulated value at sample k.

In the following evaluation, the RMSE is applied to assess both the wind estimation accuracy and the resulting state tracking performance. The wind RMSE quantifies the deviation between the estimated and actual wind components, while the state RMSE is used to evaluate the tracking error of the airship states.

Include average RMSE. (within the same category pos, vel, ... an average RMSE can be meaningful and is used throughout this thesis. (Find citation))

The remainder of this chapter is structured as follows. In Section 5.1, the performance of the wind estimation algorithm is evaluated for the considered flight and wind scenarios. Subsequently, Section 5.2 investigates the effectiveness of the proposed wind compensation strategies and their impact on the overall control performance.

5.1. Wind estimation

Following the wind estimation performance is analyzed. To use our Dryden model, which is interconnected to our airship model, we first need to specify its parameters, especially W20 as the rest is all state dependent.

ref

First of all we need to find a W20 parameter. We choose W20 = 5, which is justified in the section .

Result: Good wind estimation.... yields a good basis for wind compensation where good disturbance estimation is one requirement.

5.1.1. Sensitivity to wind parameters

As stated in introduction wind the parameter reference wind velocity W20 cannot be determined depending on states directly, it has to be estimated beforehand. We cannot know the true W20 value of the environment, but we have to give it to the wind model incorporated into the model. So we need to find an estimate for W20 that works with all wind scenarios we encounter and want to achieve good performance for (up to 30kmh)

The wind parameters W20 and wind direction cannot be measured or computed directly from the states and thus need to be estimated. In the following the sensitivity to the correct parameter is assessed. Therefore multiple tests are carried out.

First a constant parameter value for W20 was specified, creating the actual wind turbulences in the simulation. The wind estimation performance was assessed for varying estimated W20, reaching from W20 = 0 up to W20 = 30 km/h. For each estimated W20 parameter value 27 wind scenarios are run. In each wind scenario the constant wind, as well as the random seed creating the turbulences, were varied resulting in unique wind conditions for each test. For each test the RMSE between the estimated wind and actual wind are computed individually and the average RMSE along all three axis is computed. Then the results are averaged among all 27 wind scenarios. The results are shown in Table 5.1 and lead to the following conclusions.

a) Even for constant wind $W20_{\text{act}} = 0$, an estimate of $W20_{\text{est}} = 0$ leads to bad results. This can be explained when looking at the computation of the Wind Process noise, for $W20_{\text{est}} = 0$, this results in

proof

qwind = 0 , thus not allowing the wind estimate to change at all. So the initial guess will be kept forever? deviation, due to the nonperfect modelling and uncertainties in place. (For perfect modelling this deviation would be zero) b) matching W20 do not always imply best performance. Reason herefore are errors in modelling and bigger W20 more flexibility to adapt. So why not chosing W20 = 15 for all? Now look at plots showing differences and oscillations.

really? or just for constant wind = 0

The results indicate that a despite the real W20 parameter an estimated W20 = 10 yields good results for all real wind conditions created by varying W20. This can be explained as follows: a bigger W20 estimated increases the Process noise, thus giving the wind states more freedom to react to fast changing wind gusts. This comes at a cost which is a more noisy estimation of the wind. W20 estimated = 10 seems to offer a good tradeoff although bigger values may lead to an even lower RMSE. And bigger W20 leads to bigger qwind, this may have bad side-effects.

What happens with the wind estimation if W20est too big? -> if theres deviation in the measurements and the expected behaviour computed by the kalman filter, the error will more likely be attributed to the wind as its more uncertain.

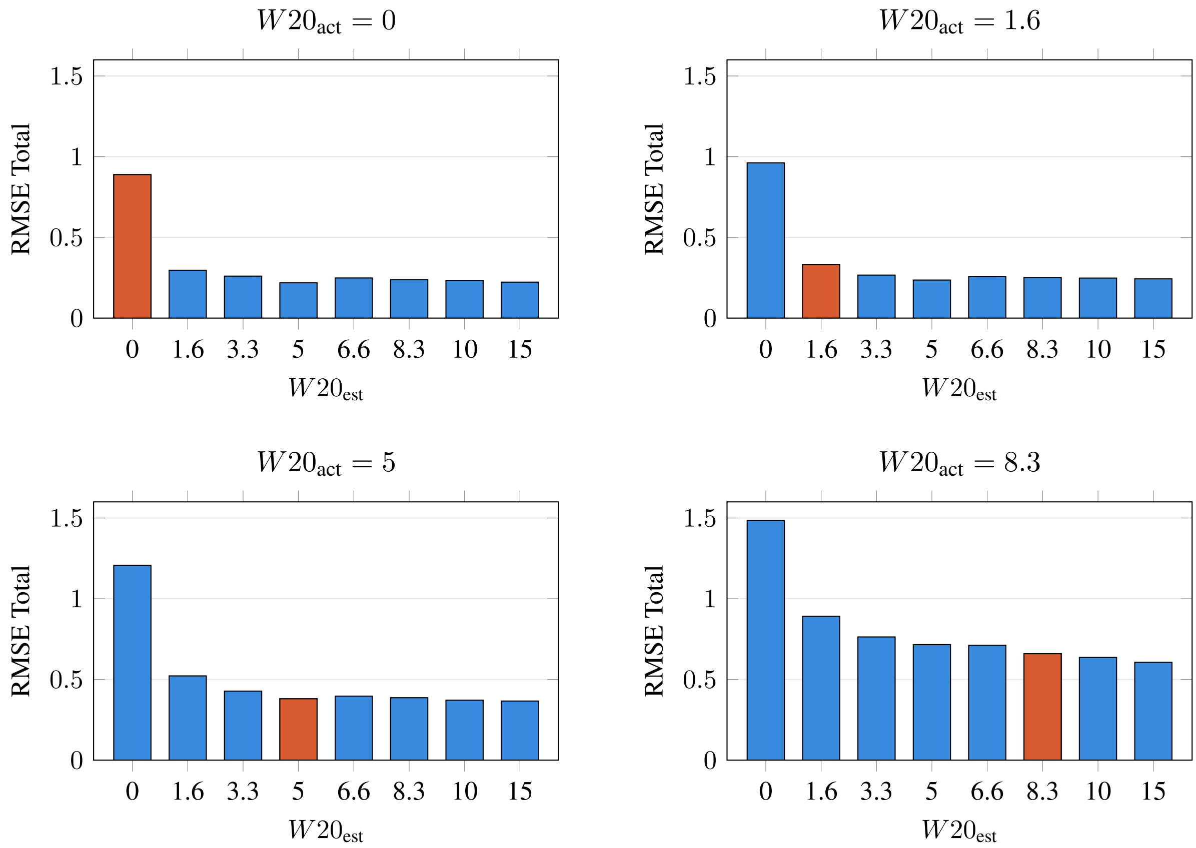


Figure 5.1.: Comparison of Total RMSE Across All Estimated $W20$ Wind Speeds. The orange bar highlights the true configuration scenario where $W20_{est} = W20_{act}$.

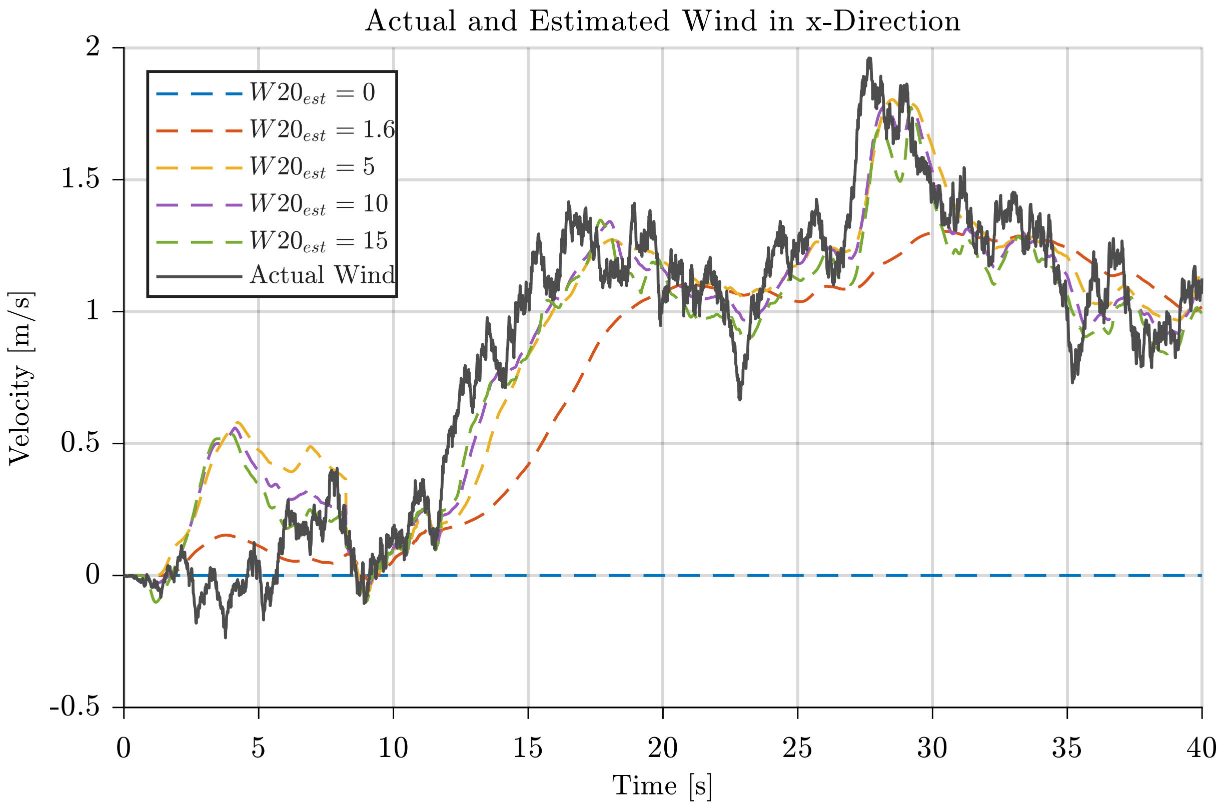


Figure 5.2.: Comparision of wind estimation performance for different estimated W20, W20 actual is 5.

5.1.2. UKF vs EKF

One can nicely see the transient pahse as described in 3.1.7. the init doesnt matter for long term

when setting the force uncertainty to zero the oscillations vanish (for R = 0 only convergence rate is improved)

Given W20 = 10, the rmse is averaged along all three wind directions.

we want to compare the performance of the UKF with that of the EKF. The UKF is fundierter while the EKF is easy to implement on the airship.

We look at converged data

Goal: Show that performance differs for W20 and flight scenario. Show that EKF and UKF perform almost equally

include in observability explanaiton

Therefore a test run was performed:

Yes actually stronger wind helps the estimate this has to do with observability

The histogram shows that the performance of EKF and UKF is very similar.

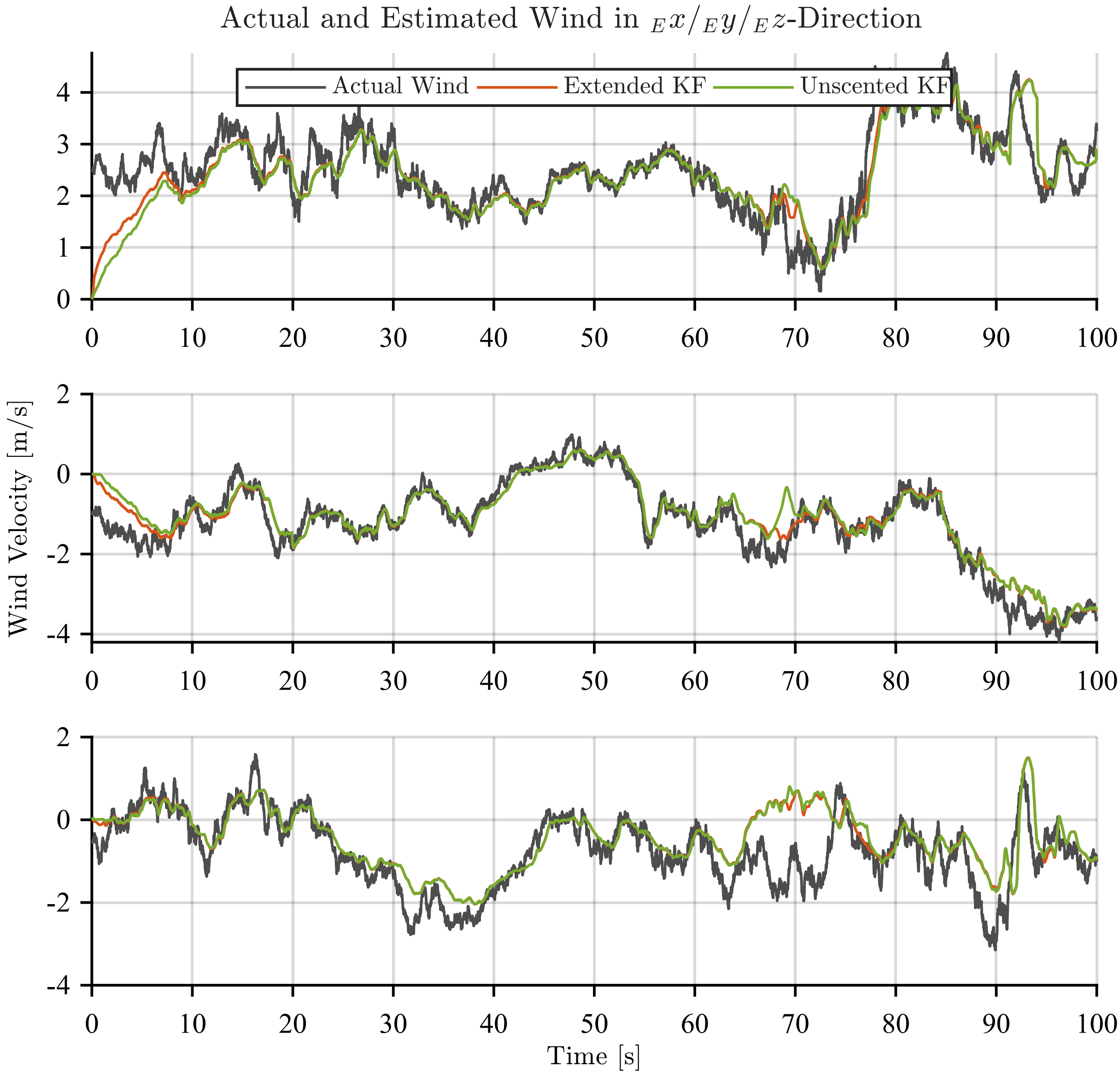


Figure 5.3.: Comparision of wind estimation performance for different estimated W20, W20 actual is 5.

Table 5.1.: Wind-estimation RMSE for the *hover* scenario by turbulence intensity W_{20}, computed on the converged window ($t = 20$–100 s).

	UKF			EKF		
W_{20}	Mean	Std	Median	Mean	Std	Median
0 (none)	0.711	0.297	0.858	0.479	0.244	0.506
5 (light)	0.612	0.389	0.609	0.459	0.232	0.485
10 (moderate)	0.493	0.212	0.432	0.476	0.171	0.457

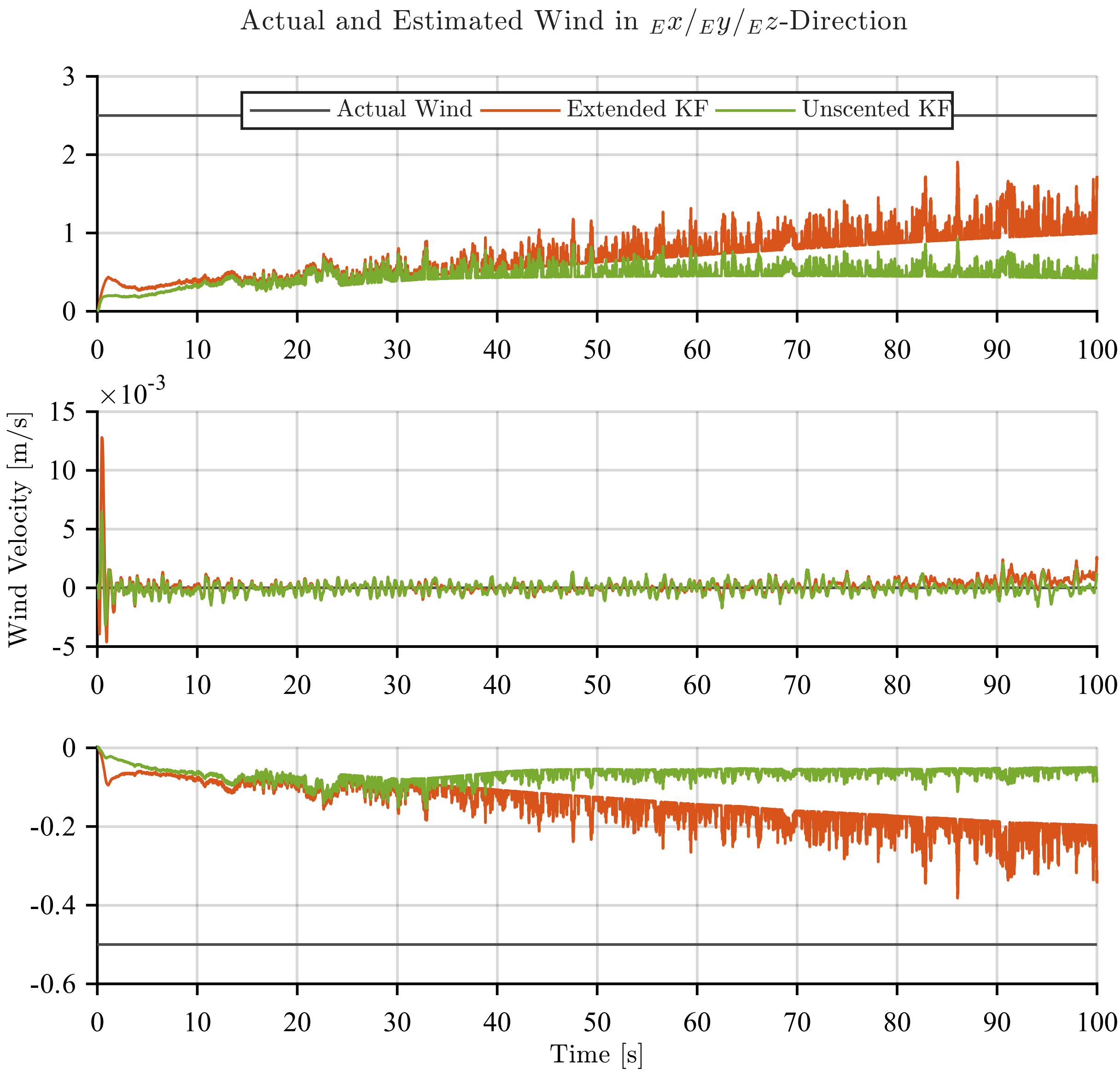


Figure 5.4.: Comparision of wind estimation performance for different estimated W20, W20 actual is 5.

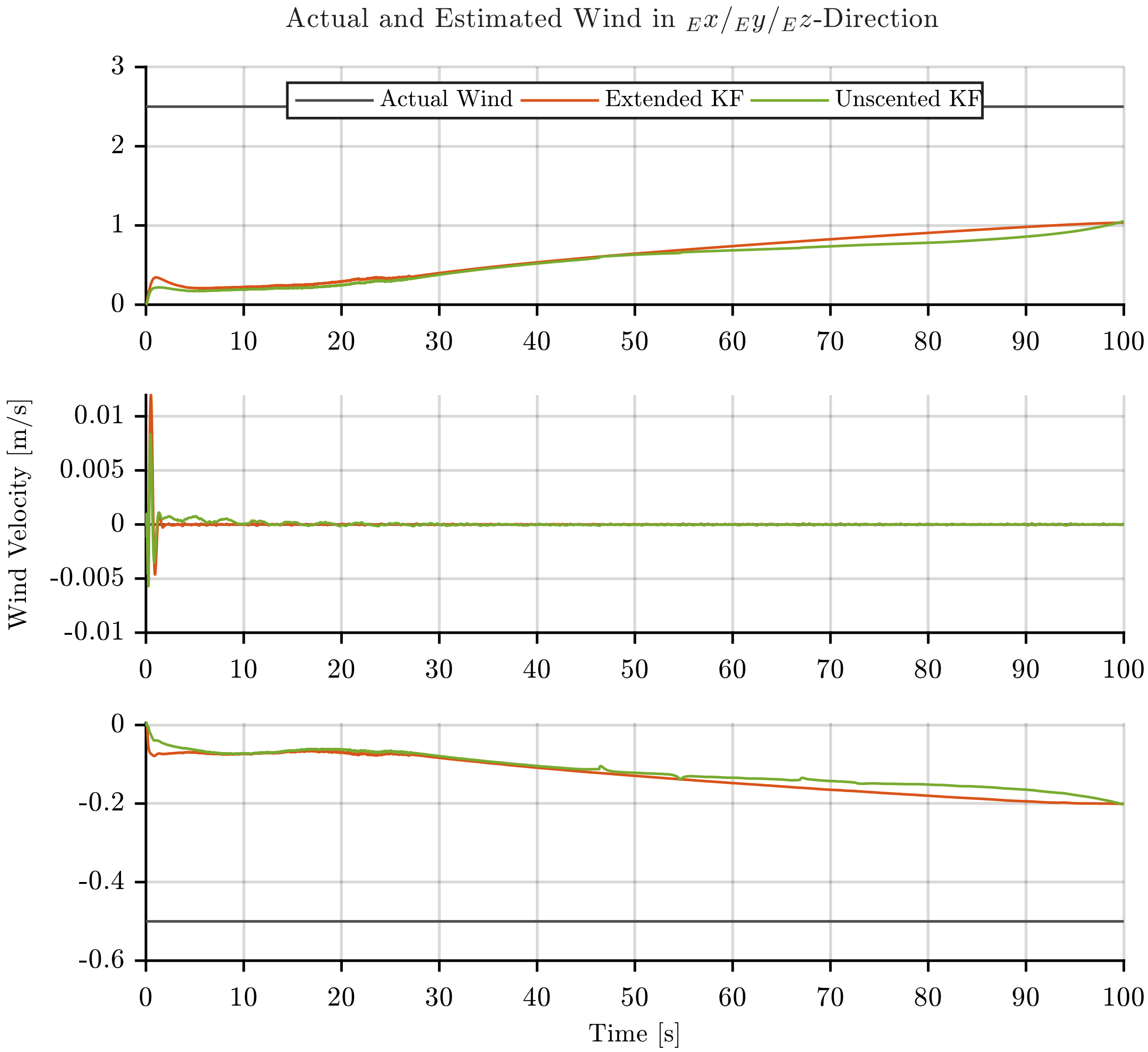


Figure 5.5.: Comparision of wind estimation performance for different estimated W20, W20 actual is 5.

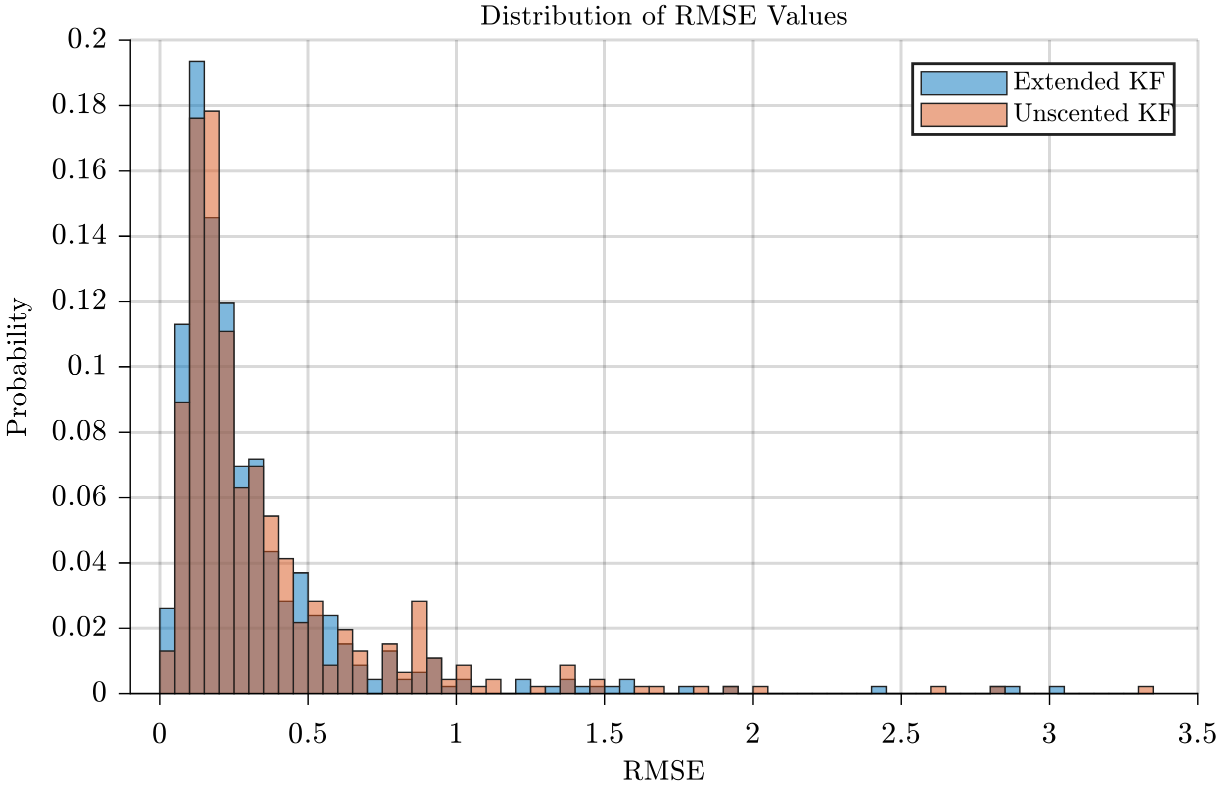
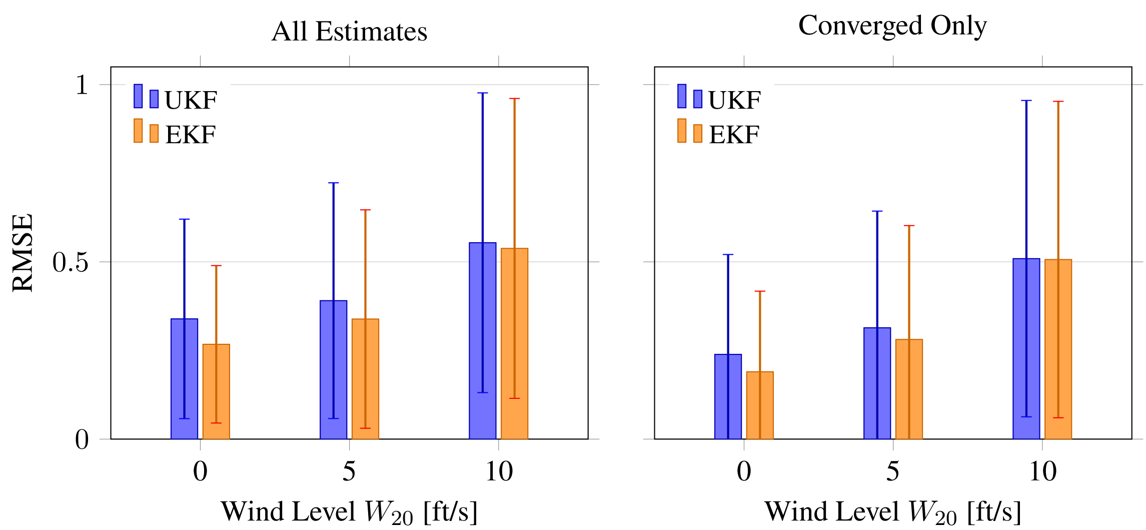


Figure 5.6.: test



the W20 barchart shows that obviously, more chellenging wind conditions, expressed by a higher wind reference value W20 result in a higher estimtion error.

The boxplot specifically show that hovering is the most challenging mode to be in, this can be because the velocity is equal to zero for this mode and theres no velocity information, which gives the wind estimation a good freedom. Find out: For hover: is the value bad for all W20 or only for constant wind?

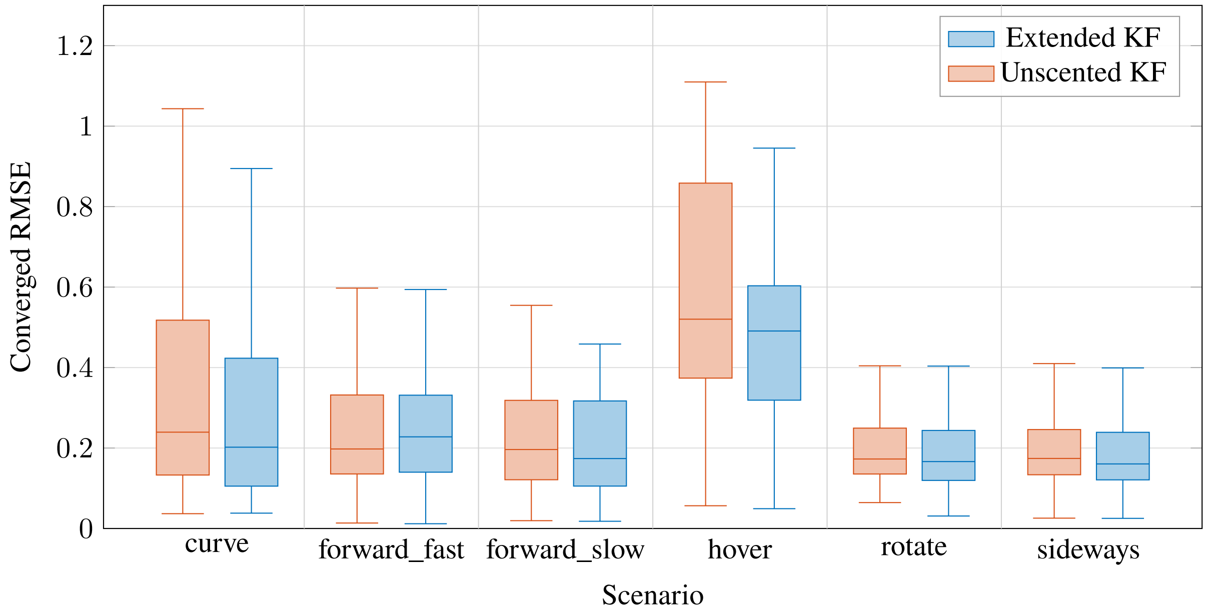


Figure 5.7.: Mean RMSE by scenario for Unscented KF and Extended KF. Boxes show the interquartile range (Q1–Q3) with the median line; whiskers extend to the most extreme point within 1.5×IQR. Outlier points beyond the whiskers are omitted.

In combination with the W20 plot that shows that the problem is not constant wind but the combination of (const wind through V = 0) and no velocity V = 0. More precisely no new wind information is delievered (compared to rotation for ex.) so a equilibrium can set with wrong wind comp. And only NDI has this problem as there we directly consider Va, whereas the DFF explicitly computes Faero with and without wind

5.1.3. Effect of Airspeed Sensor

The Airship is equipped with an 1-D airspeed sensor. Its interesting to know how the wind estimation changes if no airspeed sensor is in place, to quantify the importance of the AS.

include one time plot and one numeric comparison

5.2. Wind compensation

Compensation method has influence on the wind estimation as well.

To enable a fair comparison of the evaluated controller and gain combinations across flight scenarios, the raw RMSE values are first normalized and subsequently aggregated into a single scale-independent performance metric. This step is necessary because the twelve tracked state variables — position, velocity, attitude, and angular rate errors — differ substantially in both physical units and magnitude (e.g. position errors on the order of meters versus attitude errors on the order of radians), such that a direct average of the raw RMSE values would be dominated by the variables with the largest absolute scale.

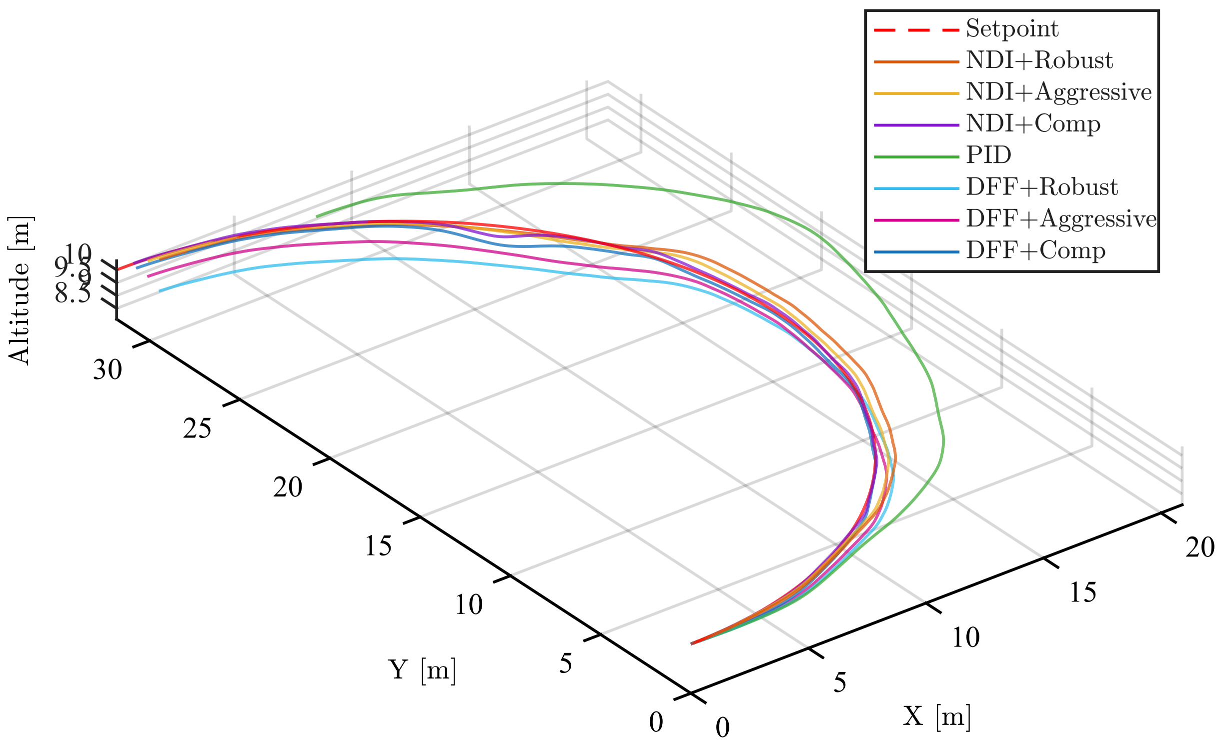


Figure 5.8.: Comparision of wind estimation performance for different estimated W20, W20 actual is 5.

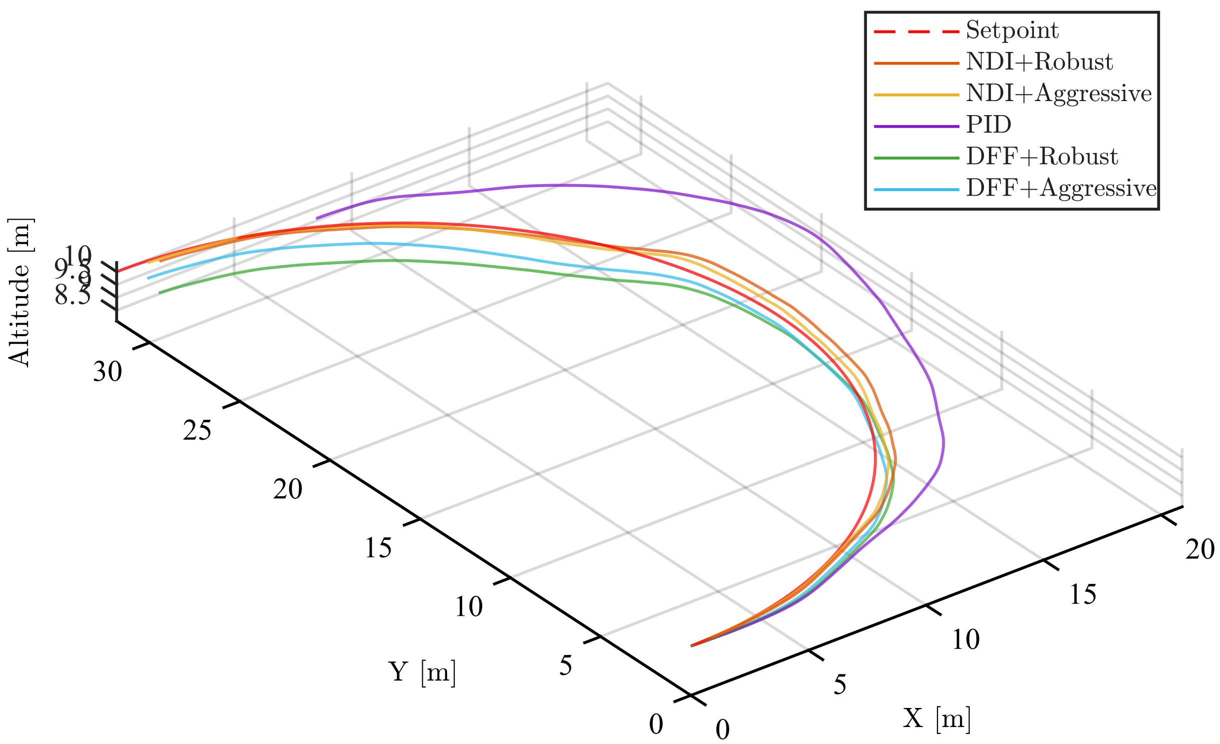


Figure 5.9.: Comparision of wind estimation performance for different estimated W20, W20 actual is 5.

Normalization. For a given scenario s, the mean RMSE of variable v is computed across all M evaluated controller/gain combinations:

$$\overline{\mathrm{RMSE}}_{s,v} = \frac{1}{M} \sum_{m=1}^{M} \mathrm{RMSE}_{s,m,v} \tag{5.2}$$

Each raw RMSE value is then expressed relative to this scenario- and variable-specific reference:

$$\widetilde{\mathrm{RMSE}}_{s,m,v} = \frac{\mathrm{RMSE}_{s,m,v}}{\overline{\mathrm{RMSE}}_{s,v}} \tag{5.3}$$

A normalized value of $\widetilde{\mathrm{RMSE}}_{s,m,v} = 1$ therefore corresponds to average performance for that scenario and variable, while values below and above 1 indicate above- and below-average estimation or control accuracy, respectively.

Aggregation. Since the normalized quantities in Eq. (5.3) are dimensionless and share a common reference (the scenario average), they can be meaningfully averaged across all $V = 12$ state variables to obtain a single figure of merit per scenario and controller/gain combination:

$$\overline{\widetilde{\mathrm{RMSE}}}_{s,m} = \frac{1}{V} \sum_{v=1}^{V} \widetilde{\mathrm{RMSE}}_{s,m,v} \tag{5.4}$$

This average normalized RMSE weights each state variable equally regardless of its native units or magnitude, and is used throughout Section **??** to compare the relative performance of the PID, NDI, and DFF control architectures under the Aggressive, Comp, and Robust gain settings.

address the hovering problem, as V = 0.

Zusammenhang zwischen schlechter windschätzung und schlechter performance im hover mode gerstellen (besonders für NDI)

one more plot showin that for W20 = 0 only hovering is bad

Table 5.2.: Average normalized RMSE per controller/gain combination, averaged across all flight scenarios and state variables.

Method	Gains	Avg. norm. RMSE
NDI	Robust	0.8309
DFF	Aggressive	0.9012
NDI	Aggressive	1.0278
DFF	Robust	1.0320
DFF	Comp	1.0405
PID	Robust	1.0643
NDI	Comp	1.1033

The wind compensation is heavily dependant on the quality of the wind estimation.

Setting:

Results We are looking for one controller which yields the best results for all scenarios. Although we may find better settings for single scenarios we do not want to need to switch.

While theoretically the compensation should allow for a more aggressive PID as the disturbance should compensate the wind disturbance this does not hold up in the results presented. Reason therefore is errors in wind estimation, convergence rates of the KF, maybe even delay of one time step, general uncertainty in the model (uncertain forces etc)

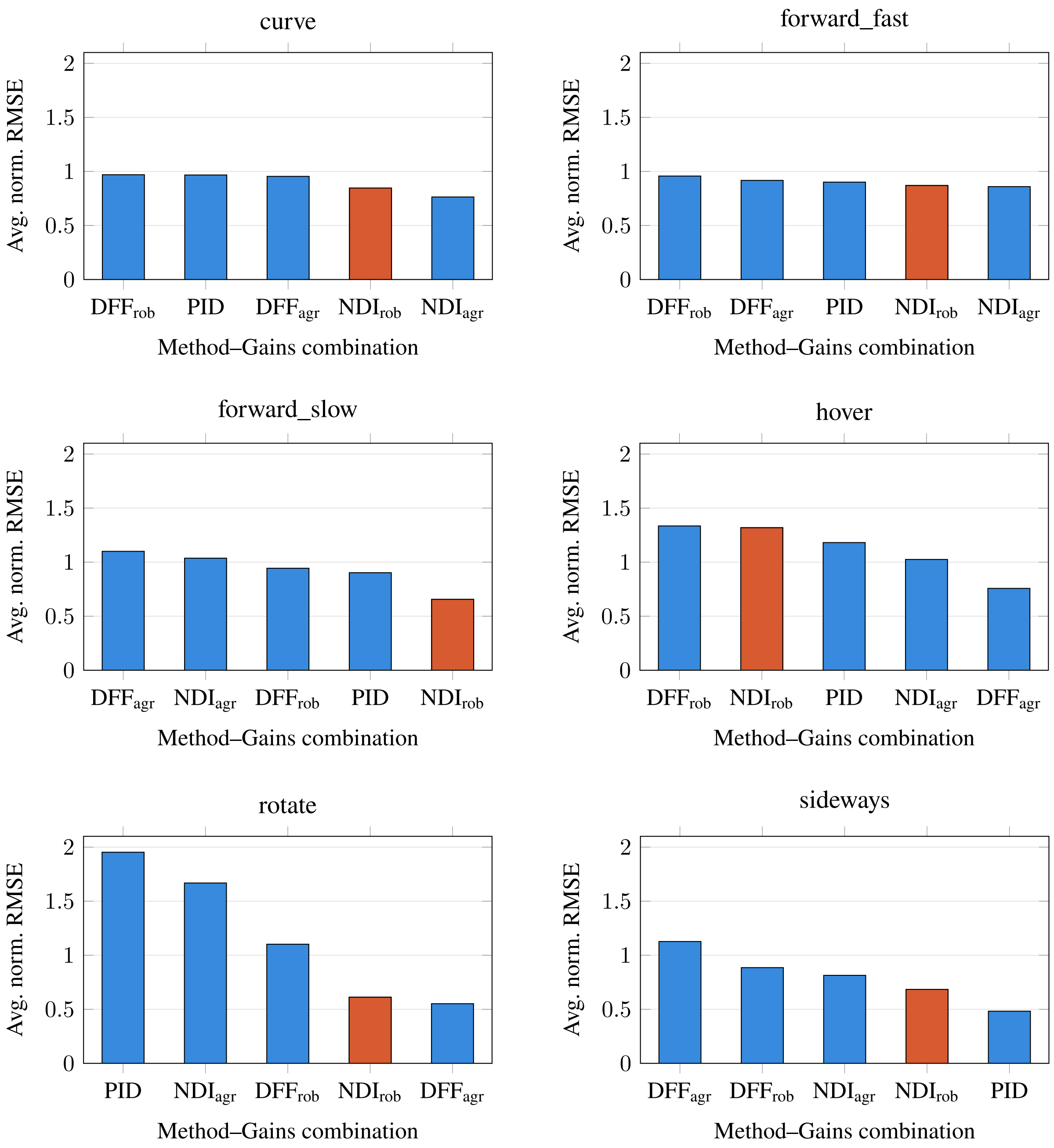


Figure 5.10.: Comparison of average normalized RMSE across selected controller/gain combinations for each flight scenario, sorted from highest to lowest. The orange bar highlights the NDI controller with Robust gains. The combinations NDI-Comp and DFF-Comp have been left out for better clarity.

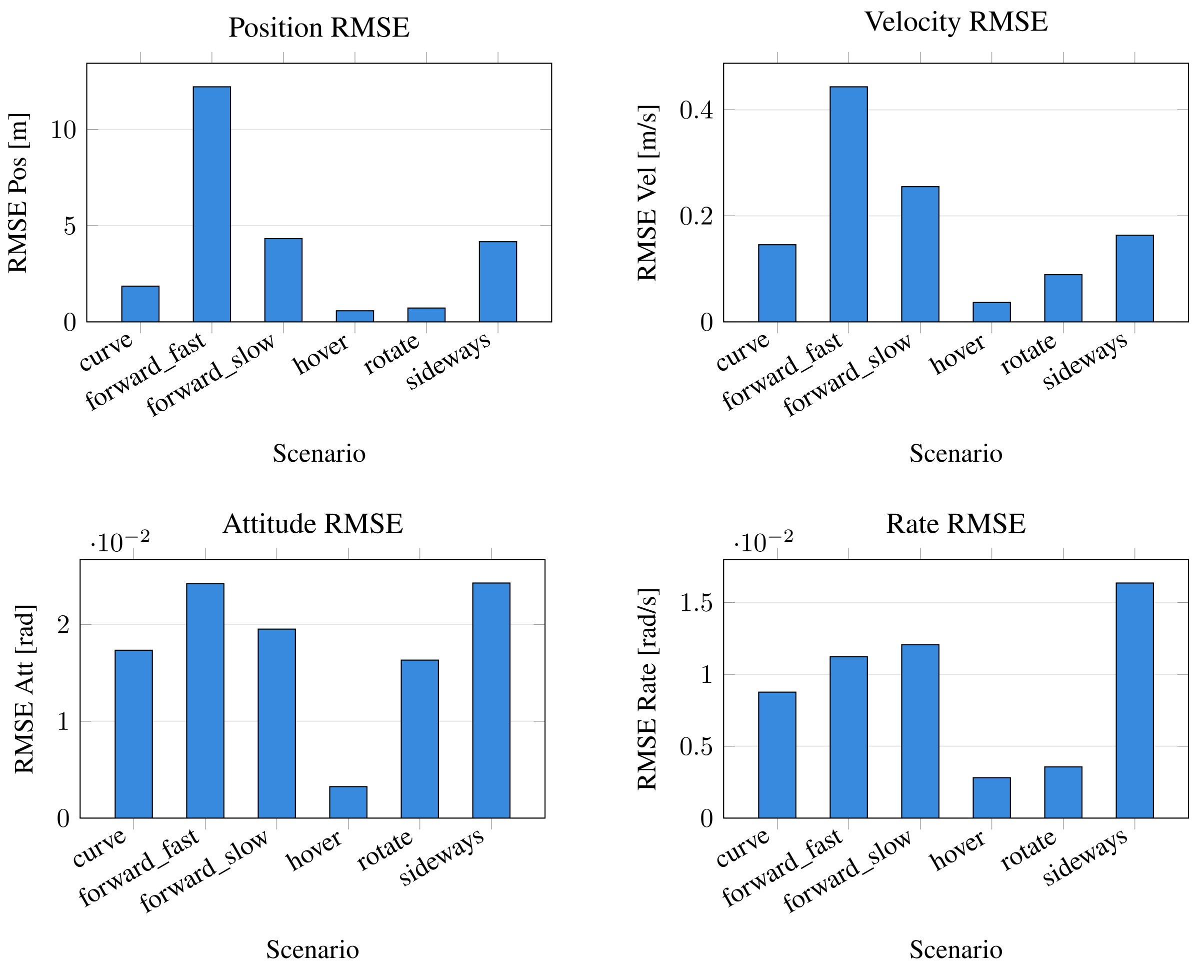


Figure 5.11.: Grouped RMSE (position, velocity, attitude, and angular rate) for the NDI controller with Robust gains across all evaluated flight scenarios.

Tuning: The PID controller is retuned on the undisturbed model, as the compensation should eliminate those, allowing for a faster PID control. It shows that the retuned PID is far too aggressive and not able to achieve good performance for all flight envelopes, although performing better on some. The reason for this is that although theoretically the Compensation should compensate for the wind, it is not able to fully cancel it due to uncertainties in the wind estimation and state measurements. Hence a robust controller is desired anyways. A combination of the robust PID with the compensation is beneficial.

Why are the Comp controllers performing bad? -> give examples when they are instable while the others are not. And state that this is the reason why they are not considered in the future.

5.2.1. Problems with Observability

Due to the lack of sensory equipment to measure airspeed the wind estimation depends heavily on the effect of the aerodynamic forces on the airship to estimate the wind velocities. Because only the aerodynamic Forces are directly dependent on the airspeed and thus on the wind velocity. The Compensation uses the current wind estimate to compute the estimation and thus requires a good estimation of wind. Exactly there lies the problem, if the compensation fully compensates for the effect of wind on the airship, the wind estimation quality automatically decreases, leading to a poor estimation and so on. To break up this cycle we need to guarantee that the information about the effect of the wind is not fully lost. To achieve this a gain, manipulating the output of the compensation is introduced. This gain in range 0 to 1 determines how aggressively the wind compensation works and balances the wind compensation with the wind estimation.

5.3. Flight Test

6. Conclusion and Outlook

Outlook

The model-based approach proves to be a powerful tool for wind estimation given a good model of the system is available. The Dryden wind model does not only provide a mathematical description for the wind speeds acting at the center of gravity but also yields a description for the effect of the wind on the angular rates, which are caused by differences in wind magnitudes within the prevailing wind field.

Including the angular rates induced by the differences in wind speeds within the wind field can improve the performance as the airship is especially vulnerable to wind fields considering the airship's large side surfaces. Angular rates of the wind are described by transition functions of order 2 (for the u-direction) and order 3 (for v, w direction) resulting in a increase of one order for every direction compared to wind speeds. It remains to investigate whether the gain in performance can justify the increased complexity.

The Wind estimation proved to be particulary reliable. This information can be used for advanced control and trajectory planning algorithms like MPC to increase the performance compared to the current PID setup.

Increase performance for hovering

Split up estimation in turbulence and mean wind to correctly use the mean wind CS for the Dryden turbulence.

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A. First Appendix

Write your appendix content here.

A.1. Parameters of the airship squid

Table A.1.: General Airship Parameters and Specifications

Parameter	**Symbol**	**Value**	**Unit**
Overall length	L	75	m
Maximum diameter	D	18	m
Envelope volume	V	10 000	m^3
Total mass	m	8 500	kg
Payload mass	m_p	1 500	kg
Helium mass	m_{He}	1 700	kg
Cruise speed	v_c	90	km/h
Maximum speed	$v_{\max}$	120	km/h
Maximum altitude	$h_{\max}$	3 000	m
Endurance	t_e	24	h
Propulsion power	P	2×150	kW
Number of engines	N_e	2	–
Buoyant force	F_b	98 000	N
Drag coefficient	C_D	0.045	–
Reference area	A	255	m^2
Moment of inertia (roll)	I_{xx}	1.2×10^5	$kg\,m^2$
Moment of inertia (pitch)	I_{yy}	8.5×10^5	$kg\,m^2$
Moment of inertia (yaw)	I_{zz}	8.9×10^5	$kg\,m^2$
Center of buoyancy offset	z_B	0.8	m

Table A.2.: Modeling Coefficients and constants

Coefficient	**Symbol**	**Value**	**Unit**
Motor Coefficient	C_{motor}	x	y
alphas	α	x	in N!!!

Table A.3.: Measurement noise standard deviations used for the Kalman filter.

Measurement	Unit	Standard deviation σ_η
Position x	m	1
Position y	m	1
Position z	m	1
Velocity u	m/s	0.15
Velocity v	m/s	0.15
Velocity w	m/s	0.25
Roll angle ϕ	°	1
Pitch angle θ	°	1
Yaw angle ψ	°	2.5
Angular velocity p	°/s	0.5
Angular velocity q	°/s	0.5
Angular velocity r	°/s	0.5

B. Second Appendix

B.1. Wind estimation

Table B.1 shows the difference in the performance for converged and non converged data.

Table B.1.: Overall wind-estimation RMSE across all test runs, for the full flight trajectory and for the converged window ($t = 20$–$100\,\mathrm{s}$).

	UKF			EKF		
	Mean	Std	Median	Mean	Std	Median
Full trajectory	0.423	0.358	0.329	0.375	0.342	0.282
Converged	0.347	0.371	0.219	0.318	0.362	0.205

Data for the wind estimation for different W20 barchart in Table B.2

Table B.2.: Wind-estimation RMSE by turbulence intensity W_{20}, computed on the converged window ($t = 20$–$100\,\mathrm{s}$).

	UKF			EKF		
W_{20}	Mean	Std	Median	Mean	Std	Median
0 (none)	0.239	0.282	0.130	0.190	0.228	0.108
5 (light)	0.314	0.329	0.200	0.281	0.321	0.192
10 (moderate)	0.509	0.446	0.363	0.506	0.446	0.355

Table B.3 shows the data for the boxplot

Table B.3.: Wind-estimation RMSE by flight scenario, computed on the converged window ($t = 20$–$100\,\mathrm{s}$).

	UKF			EKF		
Scenario	Mean	Std	Median	Mean	Std	Median
hover	0.605	0.317	0.520	0.471	0.216	0.491
forward_slow	0.276	0.304	0.196	0.258	0.302	0.174
forward_fast	0.282	0.281	0.198	0.296	0.305	0.228
sideways	0.249	0.326	0.174	0.237	0.343	0.160
curve	0.435	0.563	0.239	0.398	0.559	0.202
rotate	0.245	0.265	0.173	0.258	0.359	0.166